

Order-Sensitivity of Belief and Decision Statistics under Jeffrey Conditioning

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ABSTRACT

The effect of the order in which evidence arrives on an individual judgment has sparked much interest in both experimental and theoretical studies. Considering a case where evaluators make decisions using two correlated soft cues, the current papers asks which of their beliefs and decision statistics register a resulting effect of arrival sequence with respect to a sequence-free benchmark. With a view of implications for empirical measurement, the paper shows that an arbitrary statistic is an order of magnitude closer to the benchmark when it is insensitive to small shifts in the marginal probabilities. The paper shows how the collective marginal probabilities and the share of evaluators whose decision the sequence changes depart from the sequence-free benchmark at first order, whereas the believed association departs only at second order. Two implications follow for measurement of sequence effects through audits. First, an aggregate conforming to the benchmark is no evidence that reading sequence does not affect evaluator beliefs. Second, whether the effect of order is detectable or not depends on the kind of question being asked rather than on the number of evaluators considered.

1. Introduction

That the sequence in which evidence arrives can move the final judgment is among the oldest findings in the study of impression formation (Asch, 1946; Hogarth and Einhorn, 1992) yet the measurement of sequence dependence has rarely gathered attention. That arrival sequence of cues does effect judgment is rarely under doubt – since if individual judgments did not depend on the arrival sequence, then each cue must move belief by a factor fixed before it is received and lead to the unlikely situation where experience does not change how later cues are read at all. The sequence-dependence of Jeffrey conditioning for soft evidence in particular has attracted much debate (Diaconis and Zabell, 1982; Hawthorne, 2004). However, the question of how that dependence registers in the statistics an observer reads has received little attention. The current paper uses a two-attribute setting under Jeffrey conditioning to explore how certain statistics respond better to sequence-affected beliefs than others. It finds that the share of population affected by evidence sequence deviates from a sequence-invariant benchmark – but the believed association between attributes and the total loss due to sequence effects do not. The paper then pins down what makes an arbitrary smooth statistic become different from a sequence-free benchmark at first-order. Given that a difference from the benchmark below the observer's precision is invisible in the statistic read, the paper discusses the implications for identification of sequence effects. Despite its simplicity the two-cue setting demonstrates the central mechanism of how an attribute-local updating leaves the interactions among attributes untouched remains valid for any number of cues and attributes.

The discussions on how Jeffrey (Jeffrey, 1983) conditioning depends on arrival sequence have focused more often on whether the commutation criterion is exactly satisfied or not but not enough on how measurable its failure is in observed data. The belief-adjustment literature often models an impression as a single score (Hawthorne, 2004) or considers traits one attribute at a time (Asch, 1946). The characterisation of when two sequences lead to the same belief does consider a joint belief over attributes Diaconis and Zabell (1982), but the order dependence is more often treated as a defect to repair¹ rather than as something to measure. While changing the input from a credence that replaces the prior to a factor that multiplies it – does settle whether beliefs depend on sequence, it hardly addresses the empirical issue that an observer cannot tell whether impressions replace prior belief or adjust it.

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¹Hawthorne (2004) repairs this by letting a cue multiply the old belief instead of replacing it, and under his order-free variants the sequence effect disappears.

Whether impressions replace prior belief or adjust is a question which Hawthorne (2004), writing as a logician rather than a psychologist, leave open. To set the scope of its conclusions, the paper considers that an arrival-sequence dependence can enter through two channels, an **association** channel, in which one cue changes what the other implies and which exists only when the attributes are believed related, and a **position** channel, in which the observer weights the later cue less whatever the attributes are (Hogarth and Einhorn, 1992; Asch, 1946). In the position channel, the later cue is adopted only in part and every marginal registers the sequence even when the attributes are believed unrelated. On the other hand, when each impression is adopted in full, as in the successive updating of Diaconis and Zabell (1982), the position channel is disabled with only the association channel remaining. In the two-attribute setting with full-adoption, the paper thus considers three coordinates of a belief – two marginal probabilities and the cross-attribute association – and explains how a difference from the sequence-free benchmark below the observer’s precision is invisible in the statistic read by the observer. The question is therefore not simply whether the two sequences agree but by how much they disagree in an arbitrary statistic. The paper finds that the marginal probabilities and decisions from them carry the difference at first order in the prior covariance, while the believed association and statistics carry it only at second order. This carries clear implications for what an audit can or cannot measure about sequence dependence – an observation which may elucidate the formational mechanisms biases and prejudice.

The vignette used to discuss this setting consists of a hiring panel working through a candidate’s file on the basis of two pieces of evidence that bear on the decision—i) a credential that speaks to competence, and ii) the tone of a reference letter that speaks to trustworthiness. Both are soft evidence in the sense of Jeffrey (1983) so that neither is decisive but each leaves an impression. While the mechanism can extend to higher number of cues, we focus on the sequence dependence in a situation where one panel may read the credential first (the letter second) and an otherwise identical panel may read them in reverse. If the two attributes are believed to go together at all, the two panels finish with different beliefs about the same candidate. In the context of soft evidence the stake is real for the individual in the sense that identical evidence can license different decisions, with the difference set by arrival sequence. Apart from showing how the sequence dependence of some statistics can be invisible to an observer, the panel example also shows why that invisibility needs each impression to be adopted in full. When each impression is adopted in full, the letter fixes the belief about trustworthiness whatever the credential implied, so the sequence acts only through the believed link between the traits. As the paper shows, this means that the believed association differs between the two sequences only at second order. On the other hand, a letter adopted only in part moves that belief from wherever the credential left it (and the believed association differs at first order - see Proposition LAD). The paper therefore restricts attention to situations in which unrelated traits exhibit no sequence dependence through the position channel, and studies the measurement issues under the association channel.

The conclusions of the paper apply only when evidence is soft and when it can be assumed that impressions arrive as revised credences, not as likelihoods. Read as a level, the credential leaves the panelist in the example feeling the candidate competent, and what it delivers is a credence on the attribute’s partition into competent and not-competent categories. Read as a Bayes factor, the credential carries a likelihood ratio stating how much more probable it is under competence than in its absence, and that ratio multiplies whatever belief it meets. A Bayes factor, however, requires the probability of the same credential for a candidate who is not competent, and an impression does not supply that counterfactual. As a mechanism that holds conditionals fixed, Jeffrey updating is the unique coherent revision for credence delivered on the cue’s partition and is equivalent to making the minimal change of the prior consistent with the delivered marginal (Diaconis and Zabell, 1982)².

To avoid a clash in terminology we use “order” to describe proximity with a sequence-free Bayesian benchmark and “sequence” to denote the sequence in which evidence is read. Thus sequence effect is a sequential, myopic structure that makes Jeffrey conditioning the coherent response to a delivered credence. The distinction between first and second order on the other hand, is what an observer can or cannot see. More specifically, if the strength with which the two traits are believed to go together were represented with a prior covariance c , a departure from sequence-free benchmark is resolvable only when its size exceeds the observer’s measurement precision ϵ . For small enough c the two sequences separate, since ϵ can sit between them, $c^2 \ll \epsilon \ll c$. A first-order departure is then resolvable and a second-order one is not, so the same population can present a visible effect on one statistic and none on another. The sequence-dependence of marginals follows easily – since the likelihood a delivered credence implies must be read against the marginal in force when the cue arrives (Domotor, 1980). What is less obvious is the finding that some aggregate statistics are structurally blind to arrival order.

²See Field (1978) for how Jeffrey updates are asymmetric in the delivered-credence parametrisation and Wagner (2002) for how they commute when their inputs are held fixed in Bayes-factor form.

The key finding is presented in two parts. First, we establish that while evaluator diverges from the benchmark at first order in c (Proposition DIV), the average believed association between attributes (Proposition DEC) and the surplus-weighted loss of the population's decisions (Theorem LOS) are second order. Second, we characterise exactly which statistics can register the phenomenon at its own order. More specifically, for every interior mixture of reading sequences and across an open set of priors, a smooth statistic of the average belief is second order if and only if its differential at independence is proportional to the differential of the believed cross-attribute association (Proposition PRO). The implication for evaluation is that an audit's answer depends on which statistic is aggregated rather than on how many evaluators are averaged over. One consequence, is that the instruments most naturally trusted for detecting stereotype—belief audits asking how attributes are thought to go together—may not even be detectable.

In what follows, Section 2 sets up the model and the benchmark. Section 3 relates the paper to the literature. Section 4 establishes the individual-level results, exact immunity on a single cue and first-order divergence on two. Section 5 asks which of these carry over into the population, namely the second-order surplus-weighted loss, the decoupling of the believed cross-attribute association, the characterisation of the protected class, and the criterion split. Section 6 states the scope of results and limitations that would break the results. Section 7 discusses the key takeaways from the results. Proofs not discussed in the main text are provided in Appendix A.

2. Setup

2.1. Decision problem and surplus criterion

Two binary attributes $(A, B) \in \{0, 1\}^2$ —competence and trustworthiness, say—are each present or absent, so a belief assigns probability to the four configurations at once. It is thus a *joint law*, a single distribution over the pairs (i, j) —a point in the probability simplex Δ^3 —rather than a pair of one-attribute distributions. The two marginal distributions of the joint law record only how likely each attribute is on its own. The fixed marginals— $P(A=0) = \alpha$ and $P(B=0) = \beta$ —then leave a whole family of joint laws open—with its members differing precisely in how the attributes are believed to co-move.

Definition 1 (cross-attribute association). For a joint law Q on $\{0, 1\}^2$, the *cross-attribute association* is

$$\text{assoc}(Q) := Q_{00}Q_{11} - Q_{01}Q_{10},$$

which exists only on the joint law, never on its two marginals separately. Evaluated at the prior, $\text{assoc}(P) = c$.

The prior P is the evaluator's subjective credence, before any cue arrives, over the joint attributes of a randomly drawn candidate. It constitutes what a panelist believes about an applicant drawn at random before the file is opened, a probability for each of the four combinations (i, j) of competence and trustworthiness. It concerns the applicant's traits alone, and no outcome of the hire enters it. The prior is a point of Δ^3 fixed by three numbers, the marginals $P(A=0) = \alpha$ and $P(B=0) = \beta$ and the prior covariance c between the attributes,

$$P = \begin{pmatrix} \alpha\beta + c & \alpha(1 - \beta) - c \\ (1 - \alpha)\beta - c & (1 - \alpha)(1 - \beta) + c \end{pmatrix}, \quad \text{Cov}(A, B) = c,$$

The prior marginals α and β and the prior covariance c are the evaluator's own beliefs about a candidate drawn at random, not frequencies in the real-world. In other words, the evaluator is not estimating them and no observed rate enters. When $c = 0$ the two attributes are believed independent. The range of c is fixed by the requirement that all four cells of P stay nonnegative, which bounds it above by the smaller of $\alpha(1 - \beta)$ and $(1 - \alpha)\beta$ and below by minus the smaller of $\alpha\beta$ and $(1 - \alpha)(1 - \beta)$. These are the Fréchet–Hoeffding bounds for a 2×2 table. Simply put, the marginals $\alpha = P(A=0)$ and $\beta = P(B=0)$ denote the shares of applicants the panelist believes to lack competence and to lack trustworthiness, and the prior covariance $c = \text{assoc}(P)$ denotes how strongly the panelist believes the two traits to go together among applicants. A positive c means a competent applicant is believed the more likely to be trustworthy, and $c = 0$ means the panelist believes the traits unrelated. It is elicited by a question about both traits at once, since neither marginal carries it.

The evaluator receives two attribute-local *soft cues* so that a cue on A yields an impression (a target marginal) $q = (q_0, q_1)$ on A 's partition, with q_i the credence that $A = i$ and $q_0 + q_1 = 1$, and a cue on B yields an impression $r = (r_0, r_1)$ on B 's partition, with r_j the credence that $B = j$ and $r_0 + r_1 = 1$, and no cue bears on a joint event. An impression is thus a probability distribution on one attribute, on the same scale as the prior marginal it replaces,

$(\alpha, 1 - \alpha)$ for q and $(\beta, 1 - \beta)$ for r . In other words, there is no cue that bears on both attributes jointly and the two cues in the model are never fused into one joint event (see Assumption 3). Each impression then is a delivered credence, not a likelihood, and the product of the two impressions can be written $q \otimes r$, with $(q \otimes r)_{ij} = q_i r_j$. For the two-cues setting, a sequence $\sigma \in \{AB, BA\}$ is whether a panel read the credential or the letter first, which an observer records or an experimenter assigns, and $\lambda \in [0, 1]$ is the share of panels that read the credential first, which is counted where the sequence is recorded and unknown in a pooled audit. To compare full adoption with the belief-adjustment model of Hogarth and Einhorn (1992), in which a later impression is adopted only in part, we also consider an adoption weight $\omega \in [0, 1]$ (Section 6), which denotes how far the letter displaces what the credential had already implied about trustworthiness. At $\omega = 1$ the letter sets the rating outright, and at $\omega = 0$ the credential's implication is never moved.

The engagement decision for the evaluator is defined in terms of a desirability vector $v = (v_{ij})_{i,j \in \{0,1\}}$ —one entry per state-space cell (the same shape as P). With the score $s(P) = \langle v, P \rangle = \sum_{ij} v_{ij} P(i, j)$, the evaluator engages if and only if $s(P) \geq \tau$. This is expected-utility maximisation at the belief P —so that $s(P)$ is the expected desirability of engaging, and τ is the value of the outside option one forgoes by not engaging. Whether sequence-dependence changes what evaluators do or not is driven by the expected desirability an evaluator forgoes when their sequence-dependent belief leads them to a different action than a benchmark belief.

As we further detail in Section 2.2, the Bayes-factor benchmark P^B is used as a sequence-free reference (Wagner, 2002) to define a surplus. Writing $u := s(P^B) - \tau$ for the net surplus of engaging at the benchmark belief, the cost of taking the action that differs from the benchmark action is $|u|$. More specifically, if $u > 0$, the benchmark says engage but the evaluator doesn't and they forgo a gain of u —so that the cost is $|u|$. Likewise, when $u < 0$, the benchmark says not to engage, but the evaluator does, with the cost again being $|u|$. Whether engagement deviates from the benchmark is what feeds the aggregation picture, with τ entering the results only through $u = s(P^B) - \tau$ and never appearing on its own.

The surplus thus defined is what is of interest as the notion of loss in the population aggregate. With δ_σ as the first-order coefficient with which an evaluator's own score departs from the benchmark's under reading sequence σ (see Section 4), g as the joint density of (u, δ_σ) across the population and the sequence distribution, the expected foregone surplus is defined as

$$L(c) = \iint_{\{\text{sgn}(u) \neq \text{sgn}(u+c\delta)\}} |u| g(u, \delta) du d\delta,$$

the cost $|u|$ borne by exactly those evaluators whose decision the sequence changes, the integration variable δ ranging over the values taken by δ_σ .

2.2. The Bayes-factor benchmark

Jeffrey's rule updates on an impression by resetting the cue's marginal to the delivered credence and holding the conditionals given the cue's partition fixed, the invariance condition (Jeffrey, 2004, ch. 3), which Jeffrey (1983, ch. 11) expresses by saying that the change originates in the partition. For an impression q on A ,

$$(P_A^J P)(i, j) = q_i P(i, j) / P(A=i),$$

and symmetrically for an impression r on B . Two results of Diaconis and Zabell (1982) support the rule. The invariance condition makes the cue's partition sufficient for the prior and posterior, and minimal whenever the delivered credence differs from the prior marginal (their Theorem 2.2). The rule is also the unique minimiser of the I -divergence from the prior among all beliefs with the delivered marginal (their Theorem 5.1). The two-cue Jeffrey posteriors are the two sequences

$$P_{AB}^J = P_B^J \circ P_A^J P, \quad P_{BA}^J = P_A^J \circ P_B^J P.$$

Definition 2 (sequence effect). The *sequence effect* is the difference $P_{AB}^J - P_{BA}^J$ between the beliefs two evaluators reach from the same prior and the same two impressions, differing only in which cue they met first. It is the object of study at the individual level.

Definition 3 (gap). The *gap* is the difference between either sequence-posterior and the benchmark, $P_\sigma^J - P^B$ for $\sigma \in \{AB, BA\}$.

The benchmark P^B is not a rival updating rule but the same two impressions re-expressed—each read as a multiplicative factor relative to the prior marginal. Each impression's evidential content can be written as a Bayes factor—a likelihood matched to the cue given the prior, $\ell_i^A \propto q_i/P(A=i)$ and $\ell_j^B \propto r_j/P(B=j)$ —and, unlike a delivered credence, a Bayes factor carries the same content whatever prior it meets. Combining the two cues' Bayes-factor content is sequence-invariant (Wagner, 2002), and P^B is the single update on that combined content,

$$P^B(i, j) \propto P(i, j) \ell_i^A \ell_j^B,$$

well-defined regardless of which cue arrived first. As we later show, it coincides on any one cue with the matched Jeffrey step (Proposition IMM). We compare sequential updating on delivered credences (which does not commute) against the one-shot updating on Bayes-factor content (which does).

Example 1 (the hiring panel). Let $\alpha = \beta = \frac{1}{2}$ and $c = \frac{1}{20}$, so that the prior table is $P = \begin{pmatrix} .30 & .20 \\ .20 & .30 \end{pmatrix}$, each trait judged as likely absent as present and the two believed to go together. The credential delivers $q_0 = \frac{1}{5}$ and the letter $r_0 = \frac{7}{10}$. After the credential alone the table is $\begin{pmatrix} .12 & .08 \\ .32 & .48 \end{pmatrix}$. The A -marginal is $(.20, .80)$, as delivered, and the B -marginal has moved from $(.50, .50)$ to $(.44, .56)$ although no cue about B has been read. Reading the letter next resets the B -marginal to $(.70, .30)$ and moves the A -marginal from $.20$ to $P_{AB}^J(A=0) = .234$. In the reverse sequence the A -marginal ends at $.20$, as delivered, and the B -marginal at $P_{BA}^J(B=0) = .643$ in place of $.70$. The benchmark gives $P^B(A=0) = .227$ and $P^B(B=0) = .647$. The sequence effect is therefore $.034$ on the A -marginal and $.057$ on the B -marginal. On the believed association it is $.0002$, since $\text{assoc}(P_{AB}^J) = .0273$ and $\text{assoc}(P_{BA}^J) = .0271$, smaller than either marginal effect by a factor above 150. The odds ratio equals $9/4$ for the prior, for both sequences and for the benchmark. Against the benchmark, sequence AB departs by $.053$ on the marginal read last, by $.007$ on the marginal read first, and by $.002$ on the association.

Why a soft cue differs from a hard one. Suppose instead that the letter were a hard cue, a proposition F the panel becomes certain of, with a likelihood fixed before any belief is formed, say $P(F | B=0)/P(F | B=1) = 7/3$, and the credential likewise a proposition E with ratio $1/4$. Conditioning on E and then on F is conditioning on $E \cap F$, so the two sequences give the same belief, which is the benchmark P^B , with A -marginal $.227$ in either sequence. The soft letter delivers the credence $r_0 = \frac{7}{10}$ instead. The factor that credence implies is the ratio of the odds it delivers to the odds it meets, $7/3$ when the letter is read first against the prior marginal $\frac{1}{2}$, but $98/33$ when it is read after the credential, against the marginal $.44$ that the credential has already moved. The panel knows the credence the letter left, not what that credence would have been at another prior, and so cannot extract a portable factor from it. That dependence of the implied factor on the belief it meets is the whole of the sequence effect, and it vanishes at $c = 0$, where the credential leaves the B -marginal where it was and the two implied factors coincide.

2.3. Assumptions

With the above setup, we now state four key assumptions that limit the scope of our discussion—fixing which reading sequences, which cues, and which criterion the results apply to. The assumptions are revisited again in Section 6 to discuss how relaxing one or more of these assumptions would affect our key conclusions.

Assumption 1 (Locality in c). The prior covariance c between the two correlated attributes is treated as small, and results are derived as expansions near $c = 0$ rather than as statements holding for arbitrary c .

Assumption 2 (Soft cues). The cues are bundled, correlated, and soft, each delivering a credence on an attribute's partition, not a decisive cue.

Assumption 3 (Attribute-locality). Each impression is a credence on a single attribute's partition, never on a joint event.

Assumption 4 (Surplus criterion). Loss is measured by foregone surplus (cost $|u|$), with a threshold density that is at least integrable and with the population's desirability vectors uniformly bounded.

Before we detail what the above assumptions imply for sequence-dependent decisions in a population aggregate, we discuss how the model relates with other approaches to non-commutativity of sequence in the literature.

3. Related literature

Whether the sequence dependence of Jeffrey's probability kinematics is a problem has long been debated (Diaconis and Zabell, 1982). The dependence has been questioned on two grounds. Döring (1999) objects on a normative ground – arguing that it is unjustified in an account of rational belief change when nothing essential turns on the sequence in which experiences arrive. Hawthorne (2004) compares the usual extension of Jeffrey updating, whose full-adoption premise he calls the Amnestic Update-Factor Thesis and which the present paper adopts, with two factor models, and objects to it on psychological grounds. Judging amnestic updating to be the less plausible alternative, he turns to the normative properties of the models – declining to fully settle the objection³.

Two responses have been made to the sequence dependence that full adoption brings. One treats it as a defect of the framework. Döring (1999) argues that the sequence effect can be pronounced enough to call for an adjustment that successive Jeffrey steps cannot supply, though a single Jeffrey step from the original prior can. The other re-describes the input so that the effect disappears. Field (1978) reparametrises the update so that its input is a portable factor rather than a delivered credence, and Wagner (2002) shows that when identical learning is represented by identical Bayes factors, sequential revisions commute. The present paper takes neither of the two routes. It accepts the sequence dependence that the delivered-credence reading entails, and asks how far the two sequences disagree in each statistic of the resulting belief. Sequence dependence is not assumed here as a weight on a cue's position (position channel) but derived from a believed link between the traits (association channel). The disagreement is uneven across statistics, first order in the marginals and the share of decisions changed and second order in the believed association and the surplus-weighted loss, so whether the sequence dependence is a problem depends on which statistic an observer reads.

The ground on which the current paper adopts the premise is information-theoretic, since Jeffrey's rule is the revision that moves the prior least in I -divergence while taking on the delivered credence (Diaconis and Zabell, 1982). Under full adoption, sequence dependence sits in neither the evidence nor the evaluator alone, since it needs both cues that deliver credences rather than Bayes factors and an evaluator who believes the traits related. The current paper thus explores full-adoption only to focus on how sequence dependence may arise from believed association between attributes. For the purposes of this paper, an objection of Hawthorne's form is a reason to measure the degree of adoption, not a reason to discard the premise.

Further afield, sequence dependence has been of interest to empirical psychology at least since Asch (1946), and in economics has been taken up chiefly through the observational learning literature, though it also arises in the dynamics of discrimination (Bohren, Imas and Rosenberg, 2019). Bikhchandani, Hirshleifer and Welch (1992) and Banerjee (1992), for example, consider herd-behaviour and informational cascades in which sequence matters. In Banerjee (1992), the sequence persists because a coarse public action fails to be a sufficient statistic for private information. In Bikhchandani et al. (1992), on the other hand, a cascade is an action that conveys no private signal. The context of the paper differs significantly from that of existing Bayesian approaches in several ways. First, the object of interest in the current paper is a literal permutation non-commutativity of a fixed pair of inputs read by a single agent. Second, instead of the sequencing occurring across different agents (the sequence in which predecessors act), the current paper considers two cues read by the same agent (i.e. the sequence in which one evaluator meets two cues). Finally, the agents are assumed to be independent of each other in their actions. Instead of coupling being a feed-forward informational externality therefore—with agent n 's action entering agent $n+1$'s information set—the agents in the current model do not have any feedback. One similarity with the literature, however, is that the sequence effect in (Bikhchandani et al., 1992; Banerjee, 1992) lives on the analogue of the marginals—which remain so in the current paper as well.

Decision theory has axiomatised several alternatives to Bayesian updating, each for a single piece of news. Epstein (2006) makes the updating rule, and not only the prior, subjective, and Ortoleva (2012) axiomatises departures triggered by unexpected news. Cripps (2021) treats the sequence of signals, and his Symmetry and Divisibility axioms imply that two conditionally independent signals give the same posterior in either sequence. Read with each cue as its likelihood matched to the prior (Proposition IMM), the composite of Proposition DIV is a sequence of Bayes updates satisfying all four of his axioms, and it depends on the sequence because the second likelihood is matched to the intermediate belief, so the two sequences process different experiments. While Pettigrew and Weisberg (2025) restore sequence-invariance by pooling the prior with each new input multiplicatively before the Jeffrey step, the current paper instead accepts the non-commutativity and characterises what remains of it in aggregate. Cumulative prospect theory assigns each outcome a decision weight that depends on its rank among the outcomes of a given prospect (Tversky and Kahneman, 1992),

³“Which extension of Basic Jeffrey Updating is the more plausible model of human agents? I'm a logician, not a psychologist” (Hawthorne, 2004)

whereas in the current paper belief depends on the sequence in which two cues about one candidate are read, through the believed link between the traits, and not at all when the traits are believed unrelated. The characterisation discussed in the paper offers some clarity to the observations noted by Hogarth and Einhorn (1992). While Bayes-factor updating predicts no sequence effect anywhere (Wagner, 2002), the belief-adjustment accounts of order effects (Hogarth and Einhorn, 1992) focus on a single evaluative anchor. What the paper highlights is that such experimental analyses must also factor in the cross-attribute association – thus requiring the elicitation of a full posterior table rather than a single rating produced by step-by-step or end-of-sequence response modes.

The paper borrows the evaluator-with-binary-attribute frame from the statistical-discrimination lineage (Arrow, 1973; Coate and Loury, 1993). However, the object of interest for the current paper remains kinematic rather than an equilibrium fixed point. Unlike Coate and Loury (1993), for example, in which the stereotype is a self-confirming steady state of best responses, the current paper merely characterises the channel-free baseline. In this baseline, the differences are keyed to encounter sequence rather than to any group-marker, are individually first order, and yet remain invisible in the protected dependence statistics. It is worth highlighting that, instead of stereotypes as the selective recall of representative types (Bordalo, Coffman, Gennaioli and Shleifer, 2016), whose exaggerated associations arise across groups, the stereotype in the current paper is a believed cross-attribute association held by one evaluator and produced by coherent updating on impoverished input, not by a distortion of memory.

The audit perspective has a precedent in the economics of discrimination. Heckman (1998) showed that audit-pair estimates of discrimination rest on an assumption about unobserved productivity that “nothing guarantees” (p. 109), so that such an audit “can find discrimination when in fact none exists; it can also disguise discrimination when it is present” (p. 102), and Bohren et al. (2019) identify the source of discrimination by conditioning on the history of evaluations, since in static data, as they note, different sources generate the same patterns of observable behaviour. The obstruction here is of the same kind and arises on the belief side of an audit. The first-order signal in a protected statistic is zero in the population itself, so its silence is a failure of identification and not of estimation, which no sample size repairs, and the identifying variable is the reading sequence, which pooled data discard.

4. Which statistics register the sequence

This section asks, of each statistic of a single evaluator’s belief, whether the two reading sequences move it at first order in the prior covariance c or only at second order. Two primitive statistics carry the comparison with the sequence-free benchmark P^B of Section 2.2, the *gap*, the distance of the evaluator’s belief from P^B (Definition 3), and the *score-gap* $s(P) = \langle v, P \rangle$, the same distance read through the decision weights and set against a threshold τ . Section 4.1 shows that both are first order.

The belief statistics and the decision statistics are read from the same belief. The belief statistics are the *marginal probability* of each attribute and the *believed cross-attribute association*; the decision statistics are the surplus-weighted loss $L(c)$ and the decision *flip*, the indicator that the sequence changes the evaluator’s decision. Whether a statistic registers the sequence is settled here, for one evaluator; Section 5 shows that averaging over evaluators who met the cues in either sequence leaves the classification unchanged across an open set of priors and every interior mixture.

Before these are detailed, it is worth stating why the respective statistics were considered. For belief statistics, a two cue setting means that belief is a joint law on $\{0, 1\}^2$ and a point of Δ^3 – which in turn suggests three degrees of freedom. Thus, the marginal probability for each attribute together with a single association parameter were considered for belief statistics. The two decision statistics simply measure *how many* are affected by the arrival sequence (flip indicator) and *how much* they are affected (loss weighted by surplus $|u|$). This is the standard incidence-versus-intensity pairing of the measurement literature, where a headcount (a count of those past a threshold) and a mean shortfall (average of their distances from it) are first two members of a single parametrised family of indices (Foster, Greer and Thorbecke, 1984).

4.1. Primitives for single evaluator

Starting with the *gap*, we first state the rather trivial assertion that the difference between Jeffrey and Bayes is absent on a single cue or on two cues on independent attributes⁴.

⁴Since we focus on soft, credence-delivering evidence in the current paper, the fact that two hard cues—delivered as a likelihood—already commute, remains out of scope (see Assumption 2)

Proposition IMM (exact immunity). *On a single cue, the Jeffrey posterior equals the matched Bayesian posterior identically, for every prior and every impression. With two cues but $c = 0$ (independent attributes), $P_{AB}^J = P_{BA}^J = P^B$ exactly.*

Proof. The single cue on A means the matched likelihood $\ell_i^A = q_i Z / P(A=i)$, with Z the normaliser fixing the posterior A -marginal to q —which gives $P^B(i, j) = P(i, j) \ell_i^A / Z = q_i P(i, j) / P(A=i) = (P_A^J P)(i, j)$. For $c = 0$ the prior factorises, $P(i, j) = P(A=i)P(B=j)$. Each Jeffrey step and the Bayesian update then act on independent coordinates, so both sequences and the Bayesian posterior equal the product $q \otimes r$. $\square$

In other words, a decision turning on *one* cue is immune. Next we specify the sequence effect at the individual level—which is first order in our setup and serves as a building block for the rest of the propositions.

Proposition DIV (micro divergence). *Let $Z := \alpha\beta(1-\alpha)(1-\beta) > 0$. At $c = 0$ both reading sequences coincide with the benchmark, $P_{AB}^J = P_{BA}^J = P^B = q \otimes r$.*

(i) (gap.) *For $c \neq 0$,*

$$P_{AB}^J - P^B = c \kappa R_1 + \mathcal{O}(c^2), \quad R_1 = q \otimes (1, -1), \quad \kappa = \frac{(\alpha - q_0) r_0 (1 - r_0)}{Z},$$

$$P_{BA}^J - P^B = c \kappa' R_2 + \mathcal{O}(c^2), \quad R_2 = (1, -1) \otimes r, \quad \kappa' = \frac{(\beta - r_0) q_0 (1 - q_0)}{Z},$$

with $\kappa = 0 \iff q_0 = \alpha$ or $r_0 \in \{0, 1\}$, and symmetrically for κ' . Writing $P_\sigma^J - P^B = c D_\sigma + \mathcal{O}(c^2)$ for the leading gap direction on route σ , so that $D_{AB} = \kappa R_1$ and $D_{BA} = \kappa' R_2$, we have hence $D_\sigma \neq 0$.

(ii) (Sequence effect.) $P_{AB}^J - P_{BA}^J = c \Delta_{\text{seq}} + \mathcal{O}(c^2)$ with $\Delta_{\text{seq}} = \kappa R_1 - \kappa' R_2$. *Since R_1, R_2 are linearly independent, $\Delta_{\text{seq}} \equiv 0$ if and only if $\kappa = \kappa' = 0$; hence $\Delta_{\text{seq}} \neq 0$.*

The proof is provided in Appendix A. The mechanism behind it is straightforward. Since prior covariance c is the panel's belief about how strongly competence and trustworthiness go together, learning about one attribute is informative about the other (in the panel's own prior). A Jeffrey update on the A -cue rescales the joint distribution row by row, preserving the conditional distribution of B given A . Thus when the attributes are independent, preserving that conditional leaves B 's marginal untouched, so the two sequences trivially commute and coincide with the benchmark exactly (Proposition IMM). When $c \neq 0$, rescaling rows changes B 's marginal and the update reaches the second attribute through the covariance, so the two sequences no longer agree at first order in c .

The conditions for the first-order are not particularly stringent. Since $\kappa \propto (\alpha - q_0)$, the gap $P_{AB}^J - P^B$ vanishes when A -impression coincides with the prior A -marginal or when B -impression is degenerate ($\kappa \propto r_0(1 - r_0)$). In terms of the panel example, therefore, the gap has a nonzero leading coefficient when the credential's impression genuinely surprises ($q_0 \neq \alpha$) and the letter leaves a genuinely soft impression ($r_0(1 - r_0) > 0$).

The sequence effect (Definition 2)—i.e. the difference $P_{AB}^J - P_{BA}^J$ between the two reading sequences—is also first order for the individual evaluator. The A -first gap $P_{AB}^J - P^B$ has leading coefficient κR_1 with $R_1 = q \otimes (1, -1)$ (which holds the A -marginal fixed and moves only the B -conditional) and the B -first gap $P_{BA}^J - P^B$ has $\kappa' R_2$ with $R_2 = (1, -1) \otimes r$ (holding the B -marginal fixed). The sequence effect, being their difference, has the form $P_{AB}^J - P_{BA}^J = c(\kappa R_1 - \kappa' R_2) + \mathcal{O}(c^2)$. Because R_1 and R_2 preserve different marginals, the sequence effect is two-dimensional and does not collapse to either gap or reduce to a rescaling of one of them.

The second primitive, *score-gap* of a panel (given a fixed weight v) is the first-order coefficient with which its own score $s(P) = \langle v, P \rangle$ departs from the benchmark's. This is because for a particular route σ (AB or BA for two cues), we have

$$s(P_\sigma^J) - s(P^B) = c \delta_\sigma + \mathcal{O}(c^2), \quad \delta_\sigma := \langle v, \partial_c (P_\sigma^J - P^B)|_{c=0} \rangle.$$

4.2. Belief Statistics for single evaluator

The two belief-statistics we consider are the *believed association* between the two attributes and the *believed marginal probability* of each attribute. The evaluator panel's *believed cross-attribute association* is the association carried by the evaluator's own posterior, $\text{assoc}(P_\sigma^J)$, set against the benchmark's $\text{assoc}(P^B)$. In the example, it measures only how strongly the panel takes competence and trustworthiness to go together once the file is read (not how attributes

are observed in the candidate pool). Unlike the gap and the score gap, the *believed cross-attribute association* does not inherit the first order of Proposition DIV. As Lemma ASC shows, the immunity that is implied is exact rather than generic and it is complete for a single evaluator (before any averaging takes place).

Lemma ASC (individual-level association immunity). *For each reading sequence $\sigma \in \{AB, BA\}$ the believed association departs from the benchmark's only at second order,*

$$\text{assoc}(P_\sigma^J) - \text{assoc}(P^B) = \mathcal{O}(c^2),$$

identically in $(\alpha, \beta, q_0, r_0)$.

Proof. Since assoc (see Definition 1) is a polynomial in the cells and hence smooth, and $P_\sigma^J - P^B = c D_\sigma + \mathcal{O}(c^2)$ (Proposition DIV), the chain rule gives

$$\text{assoc}(P_\sigma^J) - \text{assoc}(P^B) = c \langle \nabla \text{assoc}(q \otimes r), D_\sigma \rangle + \mathcal{O}(c^2),$$

$$\nabla \text{assoc} \Big|_{q \otimes r} = \begin{pmatrix} q_1 r_1 & -q_1 r_0 \\ -q_0 r_1 & q_0 r_0 \end{pmatrix}.$$

With $D_{AB} = \kappa R_1$ and $D_{BA} = \kappa' R_2$ (Proposition DIV), both inner products vanish termwise:

$$\langle \nabla \text{assoc}(q \otimes r), R_1 \rangle = q_0 q_1 r_1 + q_0 q_1 r_0 - q_0 q_1 r_1 - q_0 q_1 r_0 = 0,$$

$$\langle \nabla \text{assoc}(q \otimes r), R_2 \rangle = q_1 r_0 r_1 - q_1 r_0 r_1 + q_0 r_0 r_1 - q_0 r_0 r_1 = 0,$$

so the first-order term is zero on each route. $\square$

The second belief-statistic of interest is the panel's *marginal probability* for an attribute i.e. the belief about that attribute alone, $P_\sigma^J(A=1)$. This is a linear read-out of the gap, the relevant row or column sum of $P_\sigma^J - P^B$. In audit terms, the marginal is the levels question, asking how competent evaluators take a given candidate from the pool to be and what share of evaluators would rate that candidate above a given bar. As can be seen in Proposition DIV, its rows sum to zero and its columns to $\pm\kappa$. On route *AB* the competence marginal—the attribute read *first*—therefore carries no first-order term at all, while the trustworthiness marginal—read *last*—moves at $c\kappa$. Similarly, on route *BA* we have the opposite split, rows summing to $\pm\kappa'$ and columns to zero. The marginal probability of whichever attribute is read first is thus $\mathcal{O}(c^2)$ exactly, and of whichever is read last is $\Theta(c)$ generically.

4.3. Decision Statistics for single evaluator

Panels differ not only in what they believe but also in the effects of how they act on their beliefs. The two decision statistics we're concerned with are the decision-*flip* (which becomes sequence-affected-share in the aggregate) and sequence-affected *loss*.

The panel's *flip* indicator, $\mathbb{1}\{\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)\}$, records whether the first-order perturbation $c\delta_\sigma$ of the score carries the net surplus across the threshold. At the individual level, a flip requires two things together—the perturbation $c\delta_\sigma$ must point toward the threshold, in the opposite direction from u itself, and it must be large enough to actually reach it. Formally, $\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)$ holds exactly when $\text{sgn}(c\delta_\sigma) = -\text{sgn}(u)$ and $|c\delta_\sigma| > |u|$ —equivalently, when u lies strictly between 0 and $-c\delta_\sigma$. A perturbation $c\delta_\sigma$ pushing u further from the threshold never flips it, however large c grows.

More formally, with $c^* := |u/\delta_\sigma|$ for the crossing point, a panel with $u \neq 0$ and $\delta_\sigma \neq 0$ satisfies

$$\mathbb{1}\{\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)\} = \begin{cases} 1, & |c| > c^* \wedge \text{sgn}(c\delta_\sigma) = -\text{sgn}(u), \\ 0, & \text{otherwise,} \end{cases} \quad \equiv 0 \text{ on } (-c^*, c^*),$$

and therefore

$$\frac{d^k}{dc^k} \mathbb{1}\{\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)\} \Big|_{c=0} = 0 \quad \forall k \geq 0, \quad \text{while} \quad \mathbb{1}\{\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)\} \neq 0.$$

Every Taylor coefficient at the origin vanishes even though the quantity itself does not, so it is a step function of c . As we describe in the next subsection, what the population acquires comes instead from the measure of the flip set – a band of width $\mathcal{O}(c)$ at the threshold rather than an average of individual coefficients.

$$\Pr\{\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)\} = \int_{\{|u| < |c\delta_\sigma|, \text{sgn}(u) = -\text{sgn}(c\delta_\sigma)\}} f(u) du = f(0) \cdot \mathcal{O}(c),$$

The second decision statistic is the panel's **foregone surplus**, the cost $|u|$ borne when the decision flips. It weights the flip indicator by the stake at issue, so that panels barely past the threshold contribute little and panels far from it contribute much. Writing $L_\sigma(c) := |u| \cdot \mathbb{1}\{\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)\}$ for the foregone surplus, both decision quantities share the threshold structure established above for $u \neq 0$, $\delta_\sigma \neq 0$. As the following shows, each vanishes identically until $|c|$ reaches the crossing point, and every derivative at the origin vanishes with it. Neither of the two is analytic in c and there is no expansion at the level of a single evaluator from which a leading coefficient could be read.

$$\begin{aligned} \mathbb{1}\{\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)\} &\equiv 0, & L_\sigma(c) &\equiv 0, & \text{on } (-c^*, c^*), \\ \frac{d^k}{dc^k} \mathbb{1}\{\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)\} \Big|_{c=0} &= \frac{d^k}{dc^k} L_\sigma(c) \Big|_{c=0} = 0 & \forall k \geq 0, \end{aligned}$$

5. Averaging Effects on statistics

The previous section classified statistics for one evaluator. This section shows that the classification survives averaging over a population in which a fraction $\lambda \in [0, 1]$ of evaluators meet the A -cue first (the credential in the example) and the rest the B -cue first (the letter), so that the mean belief under Jeffrey conditioning is $\bar{P}_\lambda = \lambda P_{AB}^J + (1 - \lambda) P_{BA}^J$. The population mean is a linear pool of the two posteriors, and the choice of pooling rule is immaterial at first order, since the two posteriors coincide at independence and a geometric pool of them differs from the linear one only at second order. Across an open set of priors, what makes a statistic first order or second order is the statistic itself and not the mixture λ , since the mean belief lies in the plane through P^B spanned by the leading directions of the two sequences, and a statistic's order is fixed by its differential on that plane (Propositions ORD and PRO).

Two definitions are worth pinning down before we move further. First, we define a *distortion* of an arbitrary smooth scalar statistic F as the aggregate departure from the benchmark, $F(\bar{P}) - F(P^B)$. This is the aggregate counterpart of the individual *gap* and is always a population-average object by definition. Second, we define the notion of a *protection*, which classifies that distortion by its order in c .

Definition 4 (protection). F is *protected* if its distortion is second order in the prior covariance, $F(\bar{P}) - F(P^B) = \mathcal{O}(c^2)$, and *unprotected* if that distortion is first order, $\mathcal{O}(c)$.

The distortion compares the population average with the benchmark. The same classification can be reached without either ingredient, by reading the sequence effect of Definition 2 through the statistic itself. For a smooth statistic F the sequence effect read through it is $F(P_{AB}^J) - F(P_{BA}^J)$, and it involves neither the benchmark P^B nor the mixing weight λ .

Proposition ORD (between-sequence contrast). *For any weight v ,*

$$\langle v, P_{AB}^J - P_{BA}^J \rangle = c(\kappa \langle v, R_1 \rangle - \kappa' \langle v, R_2 \rangle) + \mathcal{O}(c^2),$$

with $\kappa, \kappa', R_1, R_2$ as in Proposition DIV. The sequence effect on the A -marginal is $c\kappa' + \mathcal{O}(c^2)$ and on the B -marginal $-c\kappa + \mathcal{O}(c^2)$; each is $\Theta(c)$ generically. The sequence effect on the believed association is $\mathcal{O}(c^2)$ exactly. Across an open set of priors, a smooth statistic has a second-order sequence effect if and only if it is protected (Definition 4); both conditions hold exactly when the differential of the statistic at independence annihilates $\text{span}\{R_1, R_2\}$ (Proposition PRO below). Nothing in the display involves P^B or λ . The classification is decided by the two reading sequences, and every mixture $\bar{P}_\lambda$ inherits it.

Proof. By Proposition DIV(ii), $P_{AB}^J - P_{BA}^J = c(\kappa R_1 - \kappa' R_2) + \mathcal{O}(c^2)$; applying $\langle v, \cdot \rangle$ gives the display. With $R_1 = q \otimes (1, -1)$ and $R_2 = (1, -1) \otimes r$,

$$\langle \mathbf{1}_{A=1}, R_1 \rangle = 0, \quad \langle \mathbf{1}_{A=1}, R_2 \rangle = -1, \quad \langle \mathbf{1}_{B=1}, R_1 \rangle = -1, \quad \langle \mathbf{1}_{B=1}, R_2 \rangle = 0,$$

giving $c\kappa'$ and $-c\kappa$. For the association, $\langle \nabla \text{assoc}(q \otimes r), R_1 \rangle = \langle \nabla \text{assoc}(q \otimes r), R_2 \rangle = 0$ (proof of Proposition PRO), so its sequence effect is $\mathcal{O}(c^2)$. For the equivalence, with $M_\lambda = \lambda\kappa R_1 + (1 - \lambda)\kappa' R_2$ from Proposition DIV(i) and $\nabla F := \nabla F(q \otimes r)$,

$$\begin{aligned}\langle \nabla F, \Delta_{\text{seq}} \rangle &= \kappa \langle \nabla F, R_1 \rangle - \kappa' \langle \nabla F, R_2 \rangle, \\ \langle \nabla F, M_\lambda \rangle &= \lambda\kappa \langle \nabla F, R_1 \rangle + (1 - \lambda)\kappa' \langle \nabla F, R_2 \rangle;\end{aligned}$$

across an open set of priors κ and κ' vary independently, so either vanishes identically exactly when $\langle \nabla F, R_1 \rangle = \langle \nabla F, R_2 \rangle = 0$. $\square$

A particularly surprising finding is that for a smooth statistic of the belief, the answer is settled before any averaging over the aggregate takes place (Proposition ORD). The aggregation confines the average posterior to a two-dimensional plane, spanned by the leading directions R_1 and R_2 of the two reading sequences, so whether a statistic is protected turns only on whether its differential at independence annihilates that plane. The attribute-locality assumption and Lemma ASC ensure that the annihilator is exactly $\text{span}\{d \text{ assoc}\}$. As we summarise in Table 1, the average believed cross-attribute association (Proposition DEC) and the surplus-weighted loss $L(c)$ (Theorem LOS) are second order, whereas the aggregate gap, score gaps on unprotected weights, the marginal probabilities, and the sequence-affected share remain first order.

The obvious consequence of this property is that detectability is a property of the instrument, not of the phenomenon. To an observer of finite measurement precision ϵ , a small c allows ϵ to fall so that $c^2 \ll \epsilon \ll c$. An observer resolves a departure when the departure's size exceeds their precision ϵ and cannot resolve it when the departure sits below ϵ . Thus, the observer may resolve a first-order departure ($\epsilon \ll c$) but cannot resolve the second-order one ($c^2 \ll \epsilon$). In the panel example, two panels reading the same file in opposite sequences differ in what they believe about the candidate's trustworthiness, and near the threshold in whom they hire. A reviewer who computes the panels' average believed association between competence and trustworthiness finds it within $\mathcal{O}(c^2)$ of the benchmark and sees nothing (Lemma ASC, Proposition DEC) – whereas a reviewer computing the share of files whose decision the reading sequence changes finds a departure at $\mathcal{O}(c)$ instead. We detail the findings in the rest of the current section.

5.1. Aggregate Primitive Statistics

That both gap and score-gap remain $\Theta(c)$ in the population aggregate is somewhat trivial. This is because the aggregate simply mixes the two routes' own shifts together in proportion to λ and generically retains the sequence effects that were applicable at the individual level.

5.1.1. Gap distortion

The proof of Proposition PRO gives $\bar{P}_\lambda - P^B = c M_\lambda + \mathcal{O}(c^2)$ directly, and because R_1 and R_2 are linearly independent, M_λ vanishes at interior λ only where the coefficients of both reading sequences vanish. The aggregate gap is therefore first order on the same generic set as the gap of Proposition DIV. In other words, averaging over reading sequences reweights the leading coefficient without removing it.

5.1.2. Score gap

The average *score gap* is also a straightforward measure of the weighted score of the belief and its first-order in aggregate follows from Proposition DIV.

Lemma SCR (first-order score gap). $s(P_\sigma^J) - s(P^B) = c \delta_\sigma + \mathcal{O}(c^2)$, with δ_σ as defined in Section 4; generically $\delta_{AB} \neq \delta_{BA}$, and $\delta_\sigma = 0$ if v is constant.

Proof. Apply the linear functional $s = \langle v, \cdot \rangle$ to $P_\sigma^J - P^B = c D_\sigma + \mathcal{O}(c^2)$ from Proposition DIV. Constancy of v makes δ_σ telescope to 0 since both posteriors are normalised. $\square$

5.2. Aggregate Belief Statistics

The aggregation of belief statistics also shows that aggregation doesn't really change the sequence-effects from the individual case. As Lemma ASC shows, a particular route doesn't shift association in first-order. Mixing them does not change that. On the other hand, the case for marginal probability is not symmetrical by definition. Whichever attribute a route reads last has its marginal pinned exactly to the impression and so departs from the benchmark at first order, whereas the route reading that attribute first moves with the benchmark and its shift is zero to first order. Mixing the

Individual Quantity	Aggregate Quantity	Individual	Aggregate	Reference
<i>Primitives</i>				
Gap —gap	aggregate departure from benchmark	$\Theta(c)$ generically	$\Theta(c)$ generically	Prop. DIV; proof of Prop. PRO
Score gap —score gap on an unprotected weight, $v \notin \text{span}\{\mathbf{1}, \nabla \text{ assoc}\}$	unprotected-weight mean, $v \notin \text{span}\{\mathbf{1}, \nabla \text{ assoc}\}$	$\Theta(c)$ generically	$\Theta(c)$ generically	Lemma SCR
<i>Belief statistics</i>				
Association —believed association (odds ratio, Yule's Q)	average believed association; more generally any F with $dF \propto d \text{ assoc}$	$\mathcal{O}(c^2)$ exactly	$\mathcal{O}(c^2)$ exactly	Lemma ASC; Prop. DEC; §5.4
Marginal —marginal probability	marginal-probability drift	$\Theta(c)$ generically on the cue read last, $\mathcal{O}(c^2)$ exactly on the cue read first	$\Theta(c)$ generically for at least one attribute, every λ	Prop. DIV; Prop. DRF
<i>Decision statistics</i>				
Loss —foregone surplus	surplus-weighted loss	foregone surplus of one evaluator, 0 until $ c\delta_e $ reaches $ u $	$\mathcal{O}(c^2)$	Theorem LOS
Share —flip indicator	sequence-affected share	flip indicator for one evaluator, 0 until $ c\delta_e $ reaches $ u $, then 1	$\Theta(c)$	Prop. SHR

Table 1: Order in the prior covariance c , individual versus aggregate. $\Theta(c)$ marks a nonzero leading coefficient (genuinely first order) and $\mathcal{O}(c^2)$ a leading coefficient that vanishes (protected). Among the primitives and belief statistics, *exactly* marks a coefficient that vanishes identically in α, β, q_0, r_0 and λ , while *generically* marks one that vanishes only on a lower-dimensional set of those.

two does not zero out the first-order effect. It's not the averaging that changes the sequence, but the nature of statistic itself determines whether arrival order matters or not.

5.2.1. Aggregate cross-attribute association

Recall that the sense of stereotype in the paper depends on the cross-attribute association. With $\text{assoc}(Q) = Q_{00}Q_{11} - Q_{01}Q_{10}$ as the association of a joint law Q , the *stereotype* is the believed association absent from—or in excess of—the benchmark, a nonzero $\text{assoc}(\bar{P}) - \text{assoc}(P^B)$. By definition, stereotype is absent when the belief matches the Bayesian benchmark.

We now explain how the average believed cross-attribute association is second order because of a separable reweighting (Lemma SEP)—rather than the Jeffrey conditioning or the presence of two cues. More particularly, every updating route in the model composes to a separable reweighting—which can only rescale an association, never shift it—so the average posterior moves along a single fixed direction. Since separability holds at every N and for any rule whose steps read one attribute at a time, the protection is structural rather than a cancellation of terms and shows up as a property of attribute-locality (holding for any such rule, not Jeffrey conditioning in particular). The lemma is stated and proved for arbitrary N .

Lemma SEP (separability of attribute-local composites). *For any number N of binary attributes and a joint prior P over $\{0, 1\}^N$, any finite sequence of attribute-local updates produces a separable reweighting*

$$P'(x) = P(x) \prod_{a=1}^N g_a(x_a)$$

for some single-attribute functions g_a . Equivalently, $\log(P'/P)$ contains no interaction terms at any level.

The proof is provided in Appendix A. It proceeds by induction on the update sequence: each attribute-local step folds into that attribute's own multiplicative factor without ever coupling two coordinates, so the product form survives any finite sequence of such steps.

Proposition DEC now shows how the second-order behaviour of the distortion of the average believed association depends on the separability lemma. More specifically, Lemma SEP forces every updating route to act on the association by the pure multiplier $\text{assoc}(Q') = (g_0g_1h_0h_1/Z^2) \text{assoc}(Q)$, so all of them share the same leading factor at independence and the $\mathcal{O}(c)$ terms in $\text{assoc}(\bar{P}) - \text{assoc}(P^B)$ cancel identically. On the other hand, a non-separable updating route obeys no such identity and can shift assoc additively rather than merely rescale it, leaving a first-order distortion.

Proposition DEC (belief-level decoupling). *The distortion of the average believed association is second order—its first-order coefficient is identically zero:*

$$\text{assoc}(\bar{P}) - \text{assoc}(P^B) = \mathcal{O}(c^2).$$

Proof. Association—the $\mathcal{O}(c)$ terms coincide by separable covariance. For any rank-one (separable) reweighting $Q'_{ij} = g_i h_j Q_{ij} / Z$ of a joint law Q , with $Z = \sum_{ij} g_i h_j Q_{ij}$,

$$\text{assoc}(Q') = \frac{g_0 g_1 h_0 h_1}{Z^2} \text{assoc}(Q),$$

since the cross terms in $Q'_{00}Q'_{11} - Q'_{01}Q'_{10}$ carry the identical factor $g_0 g_1 h_0 h_1$. Explicitly,

$$\begin{aligned} Q'_{00}Q'_{11} - Q'_{01}Q'_{10} &= \frac{(g_0 h_0 Q_{00})(g_1 h_1 Q_{11}) - (g_0 h_1 Q_{01})(g_1 h_0 Q_{10})}{Z^2} \\ &= \frac{g_0 g_1 h_0 h_1}{Z^2} (Q_{00}Q_{11} - Q_{01}Q_{10}). \end{aligned}$$

Applying this with $Q = P$: by Lemma SEP, every Jeffrey route composes to exactly such a separable reweighting (at $N = 2$, a rank-one form $g_i h_j$), and the Bayes update is one directly ($g_i = q_i / \alpha_i$, $h_j = r_j / \beta_j$, writing $\alpha_0 = \alpha$, $\alpha_1 = 1 - \alpha$ and similarly for β). Every route therefore maps the prior association $\text{assoc}(P) = c$ to a *positive multiple* of c ; none

can shift it additively. Evaluating the multiplier at $c = 0$ —where all intermediate marginals are $(\alpha, 1 - \alpha)$ and $(\beta, 1 - \beta)$ regardless of route—gives the same leading factor

$$\text{assoc}(P_{AB}^J) = \text{assoc}(P_{BA}^J) = \text{assoc}(P^B) = Kc + \mathcal{O}(c^2), \quad K = \frac{q_0 q_1 r_0 r_1}{\alpha(1 - \alpha)\beta(1 - \beta)}.$$

Explicitly, at $c = 0$ one has $g_i = q_i/\alpha_i$, $h_j = r_j/\beta_j$ and $Z = 1$, so the multiplier equals $(q_0 q_1 / \alpha(1 - \alpha))(r_0 r_1 / \beta(1 - \beta)) = K$. Hence $\text{assoc}(\bar{P}) = \lambda \text{assoc}(P_{AB}^J) + (1 - \lambda) \text{assoc}(P_{BA}^J) + \mathcal{O}(c^2) = Kc + \mathcal{O}(c^2)$ (the mixture identity holds to leading order; assoc is quadratic, and the cross terms are $\mathcal{O}(c^2)$) and $\text{assoc}(\bar{P}) - \text{assoc}(P^B) = \mathcal{O}(c^2)$, the $\mathcal{O}(c)$ coefficients cancelling since Jeffrey and Bayes are both separable, so both can only rescale the prior association by the same leading factor. $\square$

The multiplier identity in the proof above yields an exact statement for the odds-ratio forms of association. In the ratio $Q'_{00}Q'_{11}/(Q'_{01}Q'_{10})$ the factor $g_0 g_1 h_0 h_1 / Z^2$ cancels, so every separable reweighting leaves the odds ratio unchanged wherever the entries are positive. It follows that P_{AB}^J , P_{BA}^J and P^B all carry the prior's odds ratio identically in c . For believed association in odds-ratio form the departure from the benchmark is exactly zero rather than $\mathcal{O}(c^2)$. Order effects live in the reset marginals, but not in the preserved association.

The proposition's implications for the example vignette are also quite revealing. In a population of panels evaluating a common pool—where each receives the same two impressions, one bearing on competence and one on trustworthiness—each panel's posterior differs from the benchmark at $\mathcal{O}(c)$ (Proposition DIV). Yet the average believed association—whether the evaluators collectively take the two attributes to go together more or less than the benchmark warrants—carries no first-order trace of the sequence (Proposition DEC). In other words, reading sequence cannot, to first order, build or erode the perceived relationship between the attributes, because both updates reweight separably, and a separable reweighting only rescales the association already present. Sequence effects therefore create no excess association or a stereotype in the above sense.

5.2.2. Aggregate Marginal-probability

The current section shows that the reading sequence does move the marginal probability in the aggregate. Proposition DRF formalises this.

Proposition DRF (marginal-probability drift). *The aggregate marginal-probability distortion is first order, with*

$$[\bar{P}(A=1) - P^B(A=1)] = c \cdot (1 - \lambda) \frac{q_0(1 - q_0)(r_0 - \beta)}{\alpha\beta(1 - \alpha)(1 - \beta)} + \mathcal{O}(c^2),$$

so for $\lambda < 1$ its sign equals $\text{sgn}(r_0 - \beta)$, sign-indefinite across cue-marginal-probability configurations. At $\lambda = 1$ this drift vanishes at first order and the B -marginal drifts with coefficient $\lambda r_0(1 - r_0)(q_0 - \alpha) / (\alpha\beta(1 - \alpha)(1 - \beta))$ with sign $\text{sgn}(q_0 - \alpha)$. For every $\lambda \in [0, 1]$, therefore, at least one marginal drifts at first order.

Proof. In the BA sequence, the A -cue is processed last, and a Jeffrey step pins its own marginal ($P_{BA}^J(A=1) = q_1$ identically in c). Similarly, in the AB sequence the later B -step pulls the A -marginal off the delivered value through the covariance so that we have

$$P_{AB}^J(A=1) = q_1 - c \frac{q_0(1 - q_0)(r_0 - \beta)}{\alpha\beta(1 - \alpha)(1 - \beta)} + \mathcal{O}(c^2),$$

Writing $K := \frac{q_0(1 - q_0)(r_0 - \beta)}{\alpha\beta(1 - \alpha)(1 - \beta)}$, the two routes give

$$P_{BA}^J(A=1) = q_1, \quad P_{AB}^J(A=1) = q_1 - cK + \mathcal{O}(c^2), \quad P^B(A=1) = q_1 - cK + \mathcal{O}(c^2),$$

Hence,

$$P_{AB}^J(A=1) - P^B(A=1) = \mathcal{O}(c^2), \quad P_{BA}^J(A=1) - P^B(A=1) = cK + \mathcal{O}(c^2).$$

Averaging over routes,

$$\begin{aligned}\bar{P}(A=1) - P^B(A=1) &= \lambda [P_{AB}^J(A=1) - P^B(A=1)] + (1 - \lambda) [P_{BA}^J(A=1) - P^B(A=1)] \\ &= \lambda \cdot \mathcal{O}(c^2) + (1 - \lambda)(cK + \mathcal{O}(c^2)) = c(1 - \lambda)K + \mathcal{O}(c^2).\end{aligned}$$

Since $q_0(1 - q_0) > 0$ and $\alpha\beta(1 - \alpha)(1 - \beta) > 0$ on the open cube, $\text{sgn } K = \text{sgn}(r_0 - \beta)$, so

$$\lambda < 1 \implies \text{sgn}[\bar{P}(A=1) - P^B(A=1)] = \text{sgn}(r_0 - \beta),$$

When $\lambda = 1$, we have

$$\lambda = 1 \implies \bar{P}(A=1) - P^B(A=1) = \mathcal{O}(c^2).$$

□

It's worth highlighting a key result that follows from the sequence affecting marginal probability at first order. The pool's average perceived competence drifts at $\mathcal{O}(c)$. In terms of the terminology we've used, the marginal probability lies outside the protected class. As a result, what an outside observer surveying the panels' joint beliefs and comparing them against the benchmark would find is that along the association axis, the average believed relationship between competence and trustworthiness would sit at essentially its benchmark value (order c^2). On the other hand, along the marginal-probability axis the average perceived marginal probability of either attribute would depart from the benchmark already at order c . It is only in the marginal probabilities that the observer could read off any trace of the sequence at all. The footprint of sequencing thus resides in marginal probabilities, not believed association.

5.3. Aggregate Decision Statistics

Both decision-statistics look at evaluators inside the threshold band who may flip their decision when the sequence changes their score. However, **Share** depends only on the band's mass, $f(0) \cdot \mathcal{O}(c)$ —thus leaving it first order. On the other hand, **Loss** weights each evaluator in that band by their stake $|u|$, and inside the band $|u|$ is itself bounded by $\mathcal{O}(c)$ from the threshold. A band of width $\mathcal{O}(c)$ weighted by a quantity of size $\mathcal{O}(c)$ leaves it second-order.

5.3.1. Surplus-weighted loss

We now show that a fixed evaluator with net surplus $u \neq 0$ is unaffected by the reading sequence until $|c\delta_\sigma|$ reaches $|u|$, so their foregone surplus is identically zero for small c rather than second order in it. Whatever sequence the population aggregate has is therefore not inherited from any member. Taking the score gap of Lemma SCR as given, the higher-order behaviour of the expected surplus loss $L(c)$ is now stated as a theorem.

Theorem LOS (surplus-weighted loss). *Let $u = s(P^B) - \tau$ have, across the population and the sequence distribution, a joint density whose u -marginal is integrable (Assumption 4). Then the expected surplus loss $L(c)$ of the Jeffrey population relative to the benchmark is $o(c)$ —strictly higher order than the individual divergence. If the u -marginal is bounded and continuous at $u = 0$ with value $f(0) > 0$, the loss is exactly second order, with constant named:*

$$L(c) = \frac{f(0)}{2} \mathbb{E}[\delta_\sigma^2 \mid u = 0] c^2 + \mathcal{O}(c^3).$$

The proof – provided in Appendix A – can be viewed as a band picture shown in Figure 1. The Jeffrey and benchmark actions differ only when the perturbation $c\delta_\sigma$ flips the sign of the surplus u , so the flip set is confined to a band of width $\mathcal{O}(c)$ at the threshold. On that band the cost $|u|$ is itself $\mathcal{O}(c)$. An $\mathcal{O}(c)$ mass of individuals, each costing $\mathcal{O}(c)$, yields an $\mathcal{O}(c^2)$ loss. More particularly, a candidate is mis-decided only when the first-order perturbation flips the sign of the surplus, $\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)$ —which in turn requires $|u| \leq |c\delta_\sigma| = \mathcal{O}(c)$. The affected evaluators are exactly those lying within $\mathcal{O}(c)$ of the threshold—the thin strip around $u = 0$ that we call the *band*. Its mass, the fraction of the population sitting in it, is $\Pr\{|u| \leq |c\delta_\sigma|\} = f(0) \cdot \mathcal{O}(c)$, since a strip of width $\mathcal{O}(c)$ carries that much probability when f is continuous at $u = 0$. The foregone surplus $|u|$ on the band is itself $\mathcal{O}(c)$ and the loss must obey

$$L(c) \leq \underbrace{\mathcal{O}(c)}_{\text{cost } |u|} \cdot \underbrace{\mathcal{O}(c)}_{\text{band mass}} = \mathcal{O}(c^2).$$

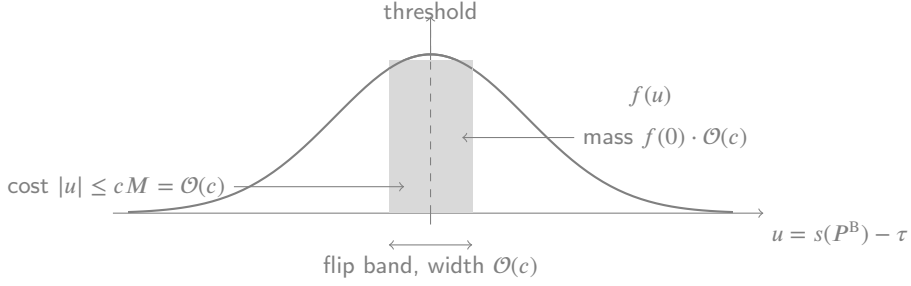


Figure 1: The band picture of Theorem LOS. Sequence-affected individuals lie in a band of width $\mathcal{O}(c)$ at the threshold; the band's mass is $f(0) \cdot \mathcal{O}(c)$, and the foregone surplus of each member is at most $cM = \mathcal{O}(c)$. The surplus-weighted loss is the product of the two factors, $\mathcal{O}(c^2)$, while the sequence-affected share returns the band's mass alone, $\mathcal{O}(c)$ (Proposition SHR).

5.3.2. Sequence-affected share—incidence versus intensity

The mechanism for the sequence-affected share is largely the same as that for surplus-weighted loss. Both integrate over the same set—the band of evaluators whose decisions the perturbation flips, of width $\mathcal{O}(c)$ at the threshold—and differ only in what they weight each member by. The surplus criterion weights by the stake $|u|$, which the band itself holds to $\mathcal{O}(c)$ whereas the 0–1 criterion weights by 1 and returns the band's mass alone.

Membership in that band is forced by a comparison of two lengths. An evaluator's distance from the threshold is $|u|$, and the reading sequence moves their score by $c\delta_\sigma$; a flip requires the second to be at least the first, so $|u| > |c\delta_\sigma|$ leaves the action unchanged whichever sequence the cues may be in. As δ_σ is bounded across the population, only evaluators within $\mathcal{O}(c)$ of the threshold can be affected at all. The condition is necessary but not sufficient—the perturbation must also point towards the threshold rather than away from it—so the affected form a subset of the band, and it is the band's mass, $f(0) \cdot \mathcal{O}(c)$, that sets the first-order rate. The following proposition formalises this.

Proposition SHR (sequence-affected share). *Suppose the u -marginal is bounded and continuous at $u = 0$ with $f(0) > 0$ (the same density hypothesis as Theorem LOS). Then under the 0–1 criterion—weight 1 for each individual whose Jeffrey and benchmark actions differ, 0 for everyone else, so that the aggregate is the incidence of sequence-affected individuals, their share of the population—the aggregate is genuinely $\mathcal{O}(c)$, not merely $o(1)$, since the band mass itself scales as $f(0) \cdot \mathcal{O}(c)$.*

Proof. Both criteria integrate over the same set, differing only in the weight attached to a member.

Step 1: the affected set. By Lemma SCR the Jeffrey score perturbs the benchmark score by $c\delta_\sigma + \mathcal{O}(c^2)$, so

$$\text{affected} \iff \text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma + \mathcal{O}(c^2)) \implies u \in S_c := \{ |u| \leq |c\delta_\sigma| + \mathcal{O}(c^2) \},$$

a band of width $\mathcal{O}(c)$ about the threshold $u = 0$, where $\text{sgn}(u)$ turns over.

Step 2: the band mass. With f bounded, continuous at 0, and $f(0) > 0$,

$$\mu(S_c) = \int_{|u| \leq |c\delta_\sigma|} f(u) du = f(0) \cdot \mathcal{O}(c),$$

first order rather than merely bounded by it, since $f(0) > 0$ and $\delta_\sigma \neq 0$ generically (Lemma SCR).

Step 3: the two criteria. On S_c the stake obeys $|u| \leq |c\delta_\sigma| = \mathcal{O}(c)$, whence

$$\underbrace{\int_{S_c} 1 d\mu}_{\text{incidence}} = \mu(S_c) = \mathcal{O}(c), \quad \underbrace{\int_{S_c} |u| d\mu}_{\text{surplus}} \leq \mathcal{O}(c) \cdot \mu(S_c) = \mathcal{O}(c^2) \quad (\text{Theorem LOS}).$$

The extra factor of c is the weight $|u|$, which the band forces to first order. □

What follows from the above proposition has strong implications for what an audit can measure. A first-order *number* of individuals are genuinely affected, each near-marginal. The second-order quantities are the surplus-weighted loss and the believed association. Therefore, the audit result cannot ever be phrased as “few are affected,” because the

incidence is first order. Whatever significance attaches to those individuals—their decisions turned on the sequence in which two impressions happened to arrive—attaches to this first-order count, not to the second-order aggregates. The two questions “how many are affected?” and “what do the aggregate dependence and surplus statistics show?” thus decouple by one order in c . Propositions SHR and PRO show that the criterion, not the phenomenon, selects which answer an aggregate instrument returns.

5.4. The protected class of aggregate statistics

We now describe the necessary and sufficient conditions under which an arbitrary smooth statistic becomes second order. We define a differential condition for a smooth statistic of the average belief—a condition whose fulfilment leaves no first-order trace of the sequence effects. More specifically, this differential condition on F is $\nabla F(q \otimes r) \in \text{span}\{J, \nabla \text{assoc}(q \otimes r)\}$ (where J is the all-ones matrix and $R_1 = q \otimes (1, -1)$, $R_2 = (1, -1) \otimes r$ are the two route directions).

Recall that *protection* means that the aggregate’s first-order coefficient vanishes identically, so the statistic sits within $\mathcal{O}(c^2)$ of its benchmark value (even though every member of the population is $\mathcal{O}(c)$ away). What we now prove is that a smooth statistic is *protected* exactly when its differential at the independent posterior $q \otimes r$ points along $d \text{ assoc}$ (the differential of the association) so that to first order the statistic moves only as the association moves. Every other smooth statistic departs at first order generically (i.e. on all but a lower-dimensional set of priors). The following proposition formalises this.

Proposition PRO (uniqueness of the protected statistic). *For $(q_0, r_0) \in (0, 1)^2$, F as C^1 in a neighbourhood of $q \otimes r$ in the probability simplex, and with the mean belief $\bar{P}_\lambda$ of Section 5.*

- (i) (Protection.) *If the differential of F at $q \otimes r$, restricted to the tangent space of the simplex, is a scalar multiple of $d \text{ assoc}$, then $F(\bar{P}_\lambda) - F(P^B) = \mathcal{O}(c^2)$ for every prior (α, β) and every λ .*
- (ii) (Uniqueness.) *Conversely, fix $\lambda \in (0, 1)$. If $F(\bar{P}_\lambda) - F(P^B) = o(c)$ for all (α, β) in some open set, then dF at $q \otimes r$, restricted to the simplex, is a multiple of $d \text{ assoc}$.*

The proof is provided in Appendix A. What drives it is that each sequence’s leading gap is a scalar multiple of a single rank-one direction— $\kappa q \otimes (1, -1)$ for one sequence and, exchanging the roles of the cues, $\kappa' (1, -1) \otimes r$ with $\kappa' = (\beta - r_0) q_0(1 - q_0)/Z$ for the other (Proposition DIV and its mirror)—so the annihilator of the two directions is two-dimensional and spanned by the constants and ∇assoc , and the converse follows from the nonconstancy of κ/κ' on any open set of priors.

The smooth statistics protected to second order are exactly the *local dependence measures*, those that, to first order at independence, depend on the posterior only through its association. This includes the association itself (Proposition DEC is the case $F = \text{assoc}$), the odds ratio, its logarithm, and Yule’s Q ; among expectations $\langle v, P \rangle$ it admits exactly those whose v lies in the two-dimensional span of the constants and ∇assoc (Lemma SCR). No statistic outside the class is second order on an open set of priors.

The quantification over λ in part (i) is what makes the class a classification of statistics rather than of populations: the aggregate first-order coefficient is $\lambda \langle \nabla F, \kappa R_1 \rangle + (1 - \lambda) \langle \nabla F, \kappa' R_2 \rangle$, and requiring it to vanish at every interior λ forces both inner products to vanish separately rather than merely cancel against each other at some particular composition of the population. Protection in the sense of Definition 4 is thus annihilation of the route directions, not cancellation between them.

In the panel example, the classification says which audit questions can register the sequencing, and the boundary falls in an instructive place. Ask how likely the panels believe a competent-seeming candidate to be trustworthy, $\bar{P}(B=1 \mid A=1)$, and the distortion is first order, since this conditional probability carries a component along the marginal probabilities and the marginal probabilities drift at $\mathcal{O}(c)$ (Proposition DRF). Ask how much *likelier* they believe a competent-seeming candidate to be trustworthy than an incompetent-seeming one, $\bar{P}(B=1 \mid A=1) - \bar{P}(B=1 \mid A=0)$, and the distortion is second order, since the subtraction removes the component along the marginal probabilities and leaves a differential at independence that is a positive multiple of $d \text{ assoc}$, placing this difference in the protected class. Stereotypes do persist, but not as trivially as is often presumed.

6. Scope

We now revisit how Assumptions 1–4 in Section 2.3 define the scope of the conclusions drawn from the model. First, a prior covariance outside the leading-order regime voids the separation (Assumption 1) that is key to much of

the propositional claim in the paper. More particularly, Assumption 1 states the results as $c \rightarrow 0$, they compare $\mathcal{O}(c)$ to $\mathcal{O}(c^2)$, a comparison that is informative only when the prior covariance c is small relative to its feasibility bound, since $\mathcal{O}(c^2)$ is then smaller than $\mathcal{O}(c)$. When c is instead a sizable fraction of that bound, the two orders are no longer numerically distinguishable, and the local-in- c results do not apply. Second, Assumption 2 states that a single cue, taken alone, produces no phenomenon at all. As Proposition IMM shows, the Jeffrey and benchmark posteriors coincide exactly on one cue and the question of sequence-dependence does not arise when Assumption 2 is not met. Third, attribute-locality (Assumption 3) makes every Jeffrey update a separable reweighting (Lemma SEP) and is the engine of Proposition DEC. A cue delivering an impression on a joint event—a credence on both attributes together rather than on a single attribute's partition, which Assumption 3 excludes—is out of scope. Such a cue reweights the joint law non-separably and could itself manufacture association from independence, so the immunity of Proposition IMM would not apply. Fourth, a non-surplus criterion breaks the surplus decoupling (Assumption 4), since the aggregate loss is first order under a fixed per-error cost (Proposition SHR). Fifth, the population modelled throughout is homogeneous in one further respect: every evaluator shares the same prior (α, β, c) , and only the encounter sequence—and, for the surplus-weighted loss, the stakes u —varies across the pool. A population whose members differ in the prior covariance c itself is outside the scope of stated results.

The two channels. Sequence dependence has two channels in this setting, and the results of Sections 4 and 5 concern one of them. One is the association channel. A cue on one attribute moves the evaluator's belief about the other through the association in the evaluator's prior, so the second cue meets a belief the first has already changed. The association channel's effect is first order in c and vanishes at independence, where each cue sets its own marginal and touches nothing else (Proposition IMM). The other is the position channel. If the cue read second is adopted only in part, then which cue is weakened depends on the sequence, and the two sequences differ even at $c = 0$, by $(1 - \omega)(\alpha - q_0)$ on the competence marginal and $-(1 - \omega)(\beta - r_0)$ on the trustworthiness marginal. The position channel is not a feature of updating on delivered credences. A Bayes-factor update that gives the second cue's factor the weight ω is sequence-dependent in the same way, the marginals differing at order zero and the believed association at first order while the odds ratio is unchanged, since commutation requires the same factor in either position (Wagner, 2002). Under full adoption the position channel is absent and the association channel is the only one, which is the setting of the results. Under partial adoption the position channel is present at $c = 0$ and moves every marginal at order zero, while the believed association still does not differ between sequences at $c = 0$ and differs at first order (Proposition LAD). The classification by order in c shifts by one order, each statistic keeping its place relative to the others, and the odds ratio is the same in both sequences at every weight. The mean belief of a population mixing the two sequences acquires a cross-product association of $-\lambda(1 - \lambda)(1 - \omega)^2(\alpha - q_0)(\beta - r_0)$ that no member holds, and the separation between the share of decisions changed and the loss is lost. What the paper says about the position channel is confined to Propositions ADJ and LAD, the first recovering its strength from the ratings and the second stating what it does to the order in c at which each statistic registers the sequence. The position channel is a weight the evaluator gives a cue for its position, present whether or not the attributes are believed related, while the association channel needs no such weight and exists only when they are.

Partial adoption. The two sequences of Sections 4 and 5 adopt each delivered credence in full. Suppose instead that the second cue is adopted only in part, so that after the first Jeffrey step the response to the second cue is a Jeffrey step on its partition to the target $(1 - \omega)m + \omega r_1$, where m is the marginal the first step left and $0 \leq \omega \leq 1$ is the adoption weight of Section 2. The rule is the averaging form of the belief-adjustment model, $S_k = (1 - w_k)S_{k-1} + w_k s(x_k)$ (Hogarth and Einhorn, 1992, Eq. 4), applied to the second cue only, which is their end-of-sequence form for two items with a constant weight, and embedded in the joint law by a Jeffrey step. Write P_{AB}^ω and P_{BA}^ω for the two sequences under this rule; $\omega = 1$ recovers P_{AB}^J and P_{BA}^J , and $\omega = 0$ ignores the second cue. Let P^A and P^B denote the belief after the A -cue alone and after the B -cue alone.

Proposition ADJ (adoption weight). *For every c and every ω , in sequence AB ,*

$$\begin{aligned} P_{AB}^\omega(B=1) - r_1 &= (1 - \omega)[P^A(B=1) - r_1], \\ P_{AB}^\omega(A=1) - q_1 &= \omega [P_{AB}^J(A=1) - q_1], \\ P_{AB}^\omega(A=1) - P_{BA}^\omega(A=1) &= \omega [P_{AB}^J(A=1) - q_1] + (1 - \omega)[q_1 - P^B(A=1)]. \end{aligned}$$

Hence the last-read marginal equals its delivered credence identically in c exactly when $\omega = 1$, the first-read marginal does so exactly when $\omega = 0$, and for $0 < \omega < 1$ neither does. The first-order coefficient in c of the first-read marginal is ω times the benchmark's coefficient for the same marginal. At $c = 0$ the third line equals $(1 - \omega)(\alpha - q_0)$, so a partially adopting evaluator shows a sequence effect at independence, where the fully adopting evaluator (Proposition IMM) and the benchmark show none. Whenever $P^A(B=1) \neq r_1$, the first line gives the weight from three marginals of one reading group, for every c ,

$$\omega = \frac{P^A(B=1) - P_{AB}^\omega(B=1)}{P^A(B=1) - r_1}.$$

The proof is provided in Appendix A.

In Example 1, an evaluator who reads the credential first rates trustworthiness at .56 before the letter, the letter alone delivers .30, and a final rating of .43 gives $\omega = (.56 - .43)/(.56 - .30) = \frac{1}{2}$.

Proposition ADJ reads the weight from the ratings. The next result says what the weight does to the order in c at which each statistic first differs between the two sequences. A statistic is *protected for every pair of cues* if its differential at every independent belief is a multiple of the differential of the association, the condition of Proposition PRO imposed at every point of the independence surface rather than at $q \otimes r$ alone; the association and the correlation coefficient are examples.

Proposition LAD (partial adoption). *Let $0 \leq \omega \leq 1$, $t_0 = (1 - \omega)\beta + \omega r_0$, $s_0 = (1 - \omega)\alpha + \omega q_0$, $t = (t_0, 1 - t_0)$, $s = (s_0, 1 - s_0)$ and $Z = \alpha\beta(1 - \alpha)(1 - \beta)$.*

(i) *At $c = 0$ the two sequences end at the independent beliefs $q \otimes t$ and $s \otimes r$, and*

$$P_{AB}^\omega - P_{BA}^\omega \Big|_{c=0} = (1 - \omega)(\beta - r_0) R_1 + (1 - \omega)(q_0 - \alpha) R_2.$$

Every marginal therefore differs between the sequences at order zero in c , the A -marginal by $(1 - \omega)(\alpha - q_0)$, while a statistic protected for every pair of cues differs between them by $\mathcal{O}(c)$.

(ii) *The believed association differs between the sequences by*

$$\text{assoc}(P_{AB}^\omega) - \text{assoc}(P_{BA}^\omega) = c \frac{(1 - \omega)H}{Z} + \mathcal{O}(c^2),$$

$$H = q_0 q_1 [\beta(1 - \beta) + \omega(\beta - r_0)^2] - r_0 r_1 [\alpha(1 - \alpha) + \omega(\alpha - q_0)^2],$$

which is first order whenever $\omega < 1$ and $H \neq 0$, and second order at $\omega = 1$.

(iii) *A statistic protected for every pair of cues differs between the sequences by $\mathcal{O}(c^2)$ for every ω if and only if its differential at every independent belief is a multiple of the differential of the log odds ratio, as for $\text{assoc}(P)/(P(A=0)P(A=1)P(B=0)P(B=1))$. The odds ratio itself is the same in both sequences for every c and every ω .*

The proof is provided in Appendix A.

Partial adoption therefore moves the sequence effect on each belief statistic other than the odds ratio one order earlier in c (Table 2), the marginals staying ahead of the association and the association ahead of the odds ratio, which never moves. Decisions do not keep their places. At $c = 0$ the benchmark is $q \otimes r$ while sequence AB ends at $q \otimes t$, so under partial adoption an evaluator's score departs from the benchmark's at order zero, and the share of decisions the sequence changes and the surplus-weighted loss are then both of order zero. The separation of Theorem LOS is a property of full adoption.

Rubric scoring, which fixes what each document delivers, tells the two settings apart, since under full adoption it leaves the sequence effect in the belief about the attribute read first and invisible in the scores (Proposition DRF), while under partial adoption the belief about the attribute read second sits $(1 - \omega)[P^A(B=1) - r_1]$ from its score (Proposition ADJ) and the two sequences differ even when the attributes are believed unrelated.

Three cautions bound these results. The classification covers this one-parameter family, not every conceivable mechanism. Asch (1946) is evidence that sequence moves marginals, six stimulus terms read in two sequences moving the proportions on eighteen response traits, and not evidence for either endpoint of the family, $\omega = 0$ or $\omega = 1$, since his own account is that early terms set a direction for the reading of later ones, so that what a later cue delivers depends

Sequence effect on	Full adoption, $\omega = 1$	Partial adoption, $\omega < 1$
Marginals	$\Theta(c)$	$\Theta(1)$
Believed association	$\mathcal{O}(c^2)$	$\Theta(c)$
Statistics agreeing with the log odds ratio to first order	$\mathcal{O}(c^2)$	$\mathcal{O}(c^2)$
Odds ratio	0	0
Share of decisions changed	$\Theta(c)$	$\Theta(1)$
Surplus-weighted loss	$\mathcal{O}(c^2)$	$\Theta(1)$

Table 2

The between-sequence effect of each statistic under full and partial adoption. Entries are orders in the prior covariance c , generic in the prior and the cues, and 0 means no sequence effect at any c . The first column is Sections 4 and 5, the second Proposition LAD; the decision rows of the second read Theorem LOS with a score gap of order zero and are not separately verified.

Question	Quantity	Answer
<i>Belief statistics</i>		
How much stereotype is carried by population's average belief?	$\text{assoc}(\bar{P}) - \text{assoc}(P^B)$	$\mathcal{O}(c^2)$ exactly (Lemma ASC; Prop. DEC)
How far does population's believed prevalence of an attribute sit from order-free inference?	$\bar{P}(A=1) - P^B(A=1)$	$\Theta(c)$ generically, at least one attribute (Prop. DRF)
<i>Decision statistics</i>		
How large is the share of candidates whose decision the reading sequence changed?	$\Pr\{\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)\}$	$\Theta(c)$ (Prop. SHR)
How large is the efficiency loss incurred through sequence-affected decisions?	$L(c) = \mathbb{E}[u \mathbb{1}\{\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)\}]$	$\mathcal{O}(c^2)$ (Theorem LOS)

Table 3

What four aggregate audits of the same population find

on its position, a mechanism outside the family here, in which each cue delivers the same credence in either position. And full adoption is a substantive commitment rather than a consequence of the level reading. Hawthorne (2004) objects that it “seems implausible that the most recent experience or non-propositional state should completely dictate belief strengths for basis sentences, with no regard for the import of previous experiences or states”. The objection applies here, and his own two-basis example shows the overwriting explicitly, so that the credential's implication for trustworthiness is erased once the letter fixes that attribute. The adoption weight is his objection stated as a parameter, and the reply offered here is not that the objection misses but that the weight is recovered from three ratings of one marginal rather than assumed.

7. Concluding remarks

The paper's central contribution is a characterisation that delineates the conditions under which an arbitrary smooth statistic of the average belief is protected, at first order, from the non-commutativity of coherent sequential updating. This characterisation is built up from the fundamental observation that the gap is not an arbitrary perturbation. Each reading sequence contributes one route direction (R_1 and R_2), so the leading-order departures of the average posterior from the benchmark are confined to the plane those two directions span. Whether a statistic is protected therefore turns on whether its differential annihilates the route plane or not.

This answer carries three implications for audits where soft-cue updating can be assumed. First, an aggregate matching the Bayesian benchmark does not license the inference that its members are Bayesian. The population's

mean believed association and its surplus-weighted loss sit within $\mathcal{O}(c^2)$ of the benchmark while every member's gap diverges at $\mathcal{O}(c)$. At leading order, a population's aggregate beliefs can look externally Bayesian—sequence-free, benchmark-conforming on the protected statistics—even when no member's belief is.

Second, what an audit can detect is decided by the statistic it computes. The protections of the two layers have separate sources—separability keeps the mean believed association first-order immune (Proposition DEC), while the stake-weighting of the flip band keeps the surplus-weighted loss second order (Theorem LOS)—and both leave the protected statistics silent about a first-order incidence of sequence-decided outcomes. A null on a protected statistic is therefore evidence about the instrument rather than about the population. Collecting more observations and disaggregating into subgroups address the estimation error, not the quantity estimated. Enlarging the sample sharpens the estimate of a protected aggregate but leaves it at $\mathcal{O}(c^2)$, and a subgroup is a subpopulation with its own mixture of reading sequences. The aggregate results are, in this sense, reassuring exactly insofar as they are cautionary. For the two-cue setting, there are no protected statistics beyond the local dependence measures and, for the threshold decision, the surplus-weighted loss.

Third, the instrumental point bears on how a stereotype is looked for outside the model. The instruments a society most naturally trusts for detecting stereotype are of the belief-auditing kind—asking what a population believes, in aggregate, about how attributes go together—and the association they measure is protected (Table 3). Coffman, Exley and Niederle (2021), for instance, ask employers how much better they believe one group of workers performs than another, and read discrimination from the hiring decisions themselves. A believed difference between groups is second order under the sequence dependence studied here, while the share of decisions the reading sequence changes is first order (Propositions PRO and SHR). Such a belief audit therefore cannot rule out a role for the reading sequence, since a population of coherent evaluators would pass it while its hiring decisions record the sequence. Debates that set improving attitudes against persistent outcome disparities may, on this reading, be comparing instruments of different asymptotic order rather than disagreeing about the same quantity.

The results discussed above locate the objection raised in the literature on sequence dependence of Jeffrey conditioning – in the levels rather than in the believed link between the traits – rather than removing the objection. While the normative question remains where Döring (1999) and Hawthorne (2004) leave it, above results bear on both objections – whether sequence dependence is pronounced or not, as Döring (1999) argues, depends on the measure, since under full adoption the reading sequence does affect the statistics. When full adoption is relaxed, as Hawthorne's objection suggests, the dependence in fact enlarges the difference, since the marginals differ at order zero and believed link at first order under partial adoption (Proposition LAD).

A. Deferred proofs

A.1. Proof of Proposition DIV

Step 1: coincidence at $c = 0$. The prior factorises, $P(i, j) = P(A=i)P(B=j)$, and each Jeffrey step resets one coordinate's marginal while leaving the other untouched, as does the matched Bayesian update. Hence

$$P_{AB}^J|_{c=0} = P_{BA}^J|_{c=0} = P^B|_{c=0} = q \otimes r, \quad (P_{AB}^J - P^B)|_{c=0} = (P_{AB}^J - P_{BA}^J)|_{c=0} = 0.$$

Step 2: Taylor expansion. Each posterior is a rational function of c whose denominator is nonvanishing near 0, so both differences are analytic there and, having no constant term by Step 1,

$$P_{AB}^J - P^B = c D + \mathcal{O}(c^2), \quad P_{AB}^J - P_{BA}^J = c \Delta_{\text{seq}} + \mathcal{O}(c^2).$$

Step 3: the leading gap. Differentiating at $c = 0$, with $Z = \alpha\beta(1-\alpha)(1-\beta) > 0$,

$$D_{00} = -D_{01} = \frac{q_0 r_0 (q_0 - \alpha)(r_0 - 1)}{Z}, \quad D_{10} = -D_{11} = \frac{-r_0 (q_0 - \alpha)(q_0 - 1)(r_0 - 1)}{Z},$$

$$\text{equivalently} \quad D = \kappa q \otimes (1, -1), \quad \kappa = \frac{(\alpha - q_0) r_0 (1 - r_0)}{Z}.$$

The $(1, -1)$ contrast makes the A -marginal preservation manifest, $D_{i0} = -D_{i1}$.

Step 4: $D \neq 0$. The scalar κ factors into the surprise in the A -impression and the spread of the B -impression,

$$\kappa \propto (\alpha - q_0), \quad \kappa \propto r_0(1 - r_0), \quad \text{so} \quad \kappa = 0 \iff q_0 = \alpha \text{ or } r_0 \in \{0, 1\},$$

and hence $D \neq 0$ for any $q_0 \neq \alpha$ in $(0, 1)$ and $r_0 \in (0, 1)$.

Step 5: the sequence effect. For the diagonal,

$$\Delta_{\text{seq}, 00} = \frac{q_0 r_0 [(\alpha - \beta) + r_0(1 - \alpha) - q_0(1 - \beta)]}{Z},$$

whose bracket vanishes only on a single hypersurface, so $\Delta_{\text{seq}, 00}$ is nonzero generically. On the symmetry locus $\{\alpha = \beta, q_0 = r_0\}$ the bracket vanishes and the effect is confined to the antisymmetric off-diagonal part,

$$\Delta_{\text{seq}, 01} = -\Delta_{\text{seq}, 10} = \frac{q_0(1 - q_0)(q_0 - \alpha)}{\alpha^2(1 - \alpha)^2} \neq 0 \quad \text{unless } q_0 = \alpha.$$

Both coefficients satisfy $\sum_k D_k = \sum_k \Delta_{\text{seq}, k} = 0$, as a difference of probability distributions must. $\square$

A.2. Proof of Lemma SEP

Write the update sequence as $P_0 = P$ and, for $t = 1, \dots, T$,

$$P_t(x) = P_{t-1}(x) h_t(x_{a_t}),$$

where a_t is the attribute the t -th update reads and h_t is a nonnegative function of that single coordinate. Jeffrey updates take exactly this form:

$$h(x_a) = \frac{q_{x_a}}{P_{t-1}(A_a = x_a)},$$

state-dependent only through its normaliser $P_{t-1}(A_a = x_a)$ —a function of x_a alone, which is all the induction below uses.

We claim, by induction on t , that $P_t(x) = P(x) \prod_{a=1}^N g_a^{(t)}(x_a)$ for some single-attribute functions $g_a^{(t)}$. The base case $t = 0$ holds with $g_a^{(0)} \equiv 1$. For the inductive step, suppose the claim at $t - 1$; then

$$P_t(x) = P(x) \left[\prod_a g_a^{(t-1)}(x_a) \right] h_t(x_{a_t}) = P(x) \prod_a g_a^{(t)}(x_a),$$

$$g_a^{(t)} := \begin{cases} g_a^{(t-1)} \cdot h_t, & a = a_t, \\ g_a^{(t-1)}, & a \neq a_t \end{cases}$$

the new factor folds into $g_{a_t}^{(t)}$ alone; no step couples two coordinates. Taking $g_a := g_a^{(T)}$ gives the displayed form.

For the equivalent formulation, on the common support (cells with $P(x) > 0$ and every accumulated factor positive) take logarithms:

$$\log \frac{P'(x)}{P(x)} = \sum_{a=1}^N \log g_a(x_a),$$

an additive function of the individual coordinates; write $\varphi := \log(P'/P)$. For any pair $a < b$, define the pairwise contrast

$$\Delta_{ab}\varphi(x) := \varphi(x) \Big|_{x_a=1, x_b=1} - \varphi(x) \Big|_{x_a=1, x_b=0} - \varphi(x) \Big|_{x_a=0, x_b=1} + \varphi(x) \Big|_{x_a=0, x_b=0},$$

holding every coordinate other than a, b fixed. Writing $R := \sum_{c \neq a, b} \log g_c(x_c)$ for the (fixed) contribution of the remaining coordinates,

$$\begin{aligned} \Delta_{ab}\varphi &= [\log g_a(1) + \log g_b(1) + R] - [\log g_a(1) + \log g_b(0) + R] \\ &\quad - [\log g_a(0) + \log g_b(1) + R] + [\log g_a(0) + \log g_b(0) + R] = 0, \end{aligned}$$

valid for every pair $a < b$ and every fixing of the other $N - 2$ coordinates. Higher-order contrasts telescope into this one: for any three coordinates a, b, c , and any φ whatsoever (additive or not),

$$\Delta_{abc}\varphi = \Delta_{ab}\varphi \Big|_{x_c=1} - \Delta_{ab}\varphi \Big|_{x_c=0},$$

and analogously at every order above three, so $\Delta_{ab}\varphi \equiv 0$ for all pairs forces every $\Delta_S\varphi \equiv 0$, $|S| \geq 2$ —exactly the interaction coefficients λ_S of the log-linear (Möbius) decomposition $\varphi = \lambda_\emptyset + \sum_a \lambda_a(x_a) + \sum_{a<b} \lambda_{ab}(x_a, x_b) + \dots$. Conversely,

$$\lambda_\emptyset := \varphi(0), \quad \lambda_a(1) := \varphi(x_a=1, x_{-a}=0) - \varphi(0), \quad \lambda_a(0) := 0$$

reproduces $\varphi(x)$ exactly at every x once every pairwise contrast vanishes, by the same telescope run backward. Separability of P'/P and vanishing interaction of order ≥ 2 are thus the same condition both ways. Nothing above is specific to Jeffrey updating, $N = 2$, or sequence length. $\square$

A.3. Proof of Theorem LOS

Step 1: bounded coefficient. The two actions differ iff $\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)$. At the fixed prior and impressions $\|\partial_c(P_\sigma^J - P^B)|_0\| < \infty$, and $\|v\| \leq V < \infty$ across the population (Assumption 4), so

$$|\delta_\sigma| = \left| \langle v, \partial_c(P_\sigma^J - P^B)|_0 \rangle \right| \leq M < \infty, \quad \sigma \in \{AB, BA\}.$$

Step 2: the flip set lies in a thin band. For fixed δ the flip set in u is the open interval strictly between 0 and $-c\delta$, whence

$$\{\text{flip}\} \subseteq B_c := \{u : 0 < |u| \leq cM\}, \quad u \in B_c \implies |u| \leq cM.$$

That the band excludes $u = 0$ does matter in the later Step 4.

Step 3: the bound. With μ the marginal law of u , Tonelli's theorem (Tao, 2011, Corollary 1.7.23) gives

$$L(c) = \iint_{\text{flip}} |u| g(u, \delta) du d\delta \leq cM \iint_{u \in B_c} g(u, \delta) du d\delta = cM \mu(B_c).$$

Step 4: the band is null in the limit. The bands nest and exhaust,

$$c' < c \implies B_{c'} \subseteq B_c, \quad \bigcap_{c>0} B_c = \emptyset,$$

since $u \neq 0$ exits once $cM < |u|$ and $u = 0$ lies in no B_c . As μ is finite, continuity from above (Tao, 2011, §1.4) yields

$$\mu(B_c) \downarrow \mu(\emptyset) = 0, \quad \frac{L(c)}{c} \leq M \mu(B_c) \longrightarrow 0, \quad L(c) = o(c),$$

for any integrable density, and irrespective of an atom at $u = 0$, where $|u| = 0$.

Step 5: named constant available under continuity. Let the u -marginal be continuous at 0 with $f(0) > 0$. For fixed δ ,

$$\int_0^{|\delta|} t g(0, \delta) dt = \frac{1}{2} g(0, \delta) \delta^2 c^2 + \mathcal{O}(c^3),$$

and with

$$f(0) = \int g(0, \delta) d\delta, \quad \mathbb{E}[\delta^2 \mid u = 0] = \frac{1}{f(0)} \int \delta^2 g(0, \delta) d\delta,$$

integration over δ gives

$$L(c) = \frac{f(0)}{2} \mathbb{E}[\delta^2 \mid u = 0] c^2 + \mathcal{O}(c^3).$$

$\square$

A.4. Proof of Proposition PRO

Step 1: the aggregate moves in a plane. By Proposition DIV and its mirror,

$$P_{AB}^J - P^B = c \kappa R_1 + \mathcal{O}(c^2), \quad P_{BA}^J - P^B = c \kappa' R_2 + \mathcal{O}(c^2),$$

$$R_1 = q \otimes (1, -1), \quad \kappa = \frac{(\alpha - q_0)r_0(1 - r_0)}{Z}, \quad R_2 = (1, -1) \otimes r, \quad \kappa' = \frac{(\beta - r_0)q_0(1 - q_0)}{Z},$$

so with $M_\lambda := \lambda \kappa R_1 + (1 - \lambda) \kappa' R_2$ the chain rule gives

$$\bar{P}_\lambda - P^B = c M_\lambda + \mathcal{O}(c^2), \quad F(\bar{P}_\lambda) - F(P^B) = c \langle \nabla F(q \otimes r), M_\lambda \rangle + \mathcal{O}(c^2).$$

Step 2: the annihilator is span{J, ∇ assoc}. The directions are independent— $aR_1 + bR_2 = 0$ gives $b(r_0 + r_1) = b = 0$ on adding the top-row entries, then $aq_0 = 0$ and $a = 0$ —so

$$\dim\{V \in \mathbb{R}^{2 \times 2} : \langle V, R_1 \rangle = \langle V, R_2 \rangle = 0\} = 4 - 2 = 2.$$

Both J (all entries 1) and $\nabla \text{assoc}(q \otimes r)$ lie in this space,

$$\langle J, R_1 \rangle = \langle J, R_2 \rangle = 0, \quad \langle \nabla \text{assoc}, R_1 \rangle = \langle \nabla \text{assoc}, R_2 \rangle = 0 \quad (\text{Lemma ASC}),$$

and are linearly independent, hence span it.

Step 3: part (i). If $dF|_{q \otimes r}$ restricted to the simplex is a multiple of $d \text{ assoc}$, then

$$\nabla F(q \otimes r) = xJ + y \nabla \text{assoc}(q \otimes r) \implies \langle \nabla F, M_\lambda \rangle = 0 \quad \text{identically in } (\alpha, \beta, \lambda).$$

Step 4: part (ii). Put $x = \langle \nabla F, R_1 \rangle$ and $y = \langle \nabla F, R_2 \rangle$, so the first-order coefficient $\lambda \kappa x + (1 - \lambda) \kappa' y$ vanishes on an open set U of priors. Suppose $y \neq 0$. If $x = 0$ then $\kappa' y \equiv 0$ on U , and κ' vanishes only on $\{\beta = r_0\}$, forcing $y = 0$; so $x \neq 0$ and, on $U \setminus \{\beta = r_0\}$,

$$\frac{\kappa}{\kappa'} = -\frac{(1 - \lambda)y}{\lambda x} = \text{const}, \quad \text{while} \quad \frac{\kappa}{\kappa'} = \frac{(\alpha - q_0)r_0(1 - r_0)}{(\beta - r_0)q_0(1 - q_0)},$$

$$\frac{\partial}{\partial \alpha} \frac{\kappa}{\kappa'} = \frac{r_0(1 - r_0)}{(\beta - r_0)q_0(1 - q_0)} \neq 0 \quad \text{wherever } \beta \neq r_0,$$

a contradiction. Hence $y = 0$, then $\kappa x \equiv 0$ on U forces $x = 0$, so $\nabla F(q \otimes r) \in \text{span}\{J, \nabla \text{assoc}\}$. On the tangent space of the simplex, whose vectors have coordinates summing to zero, J acts as zero, so dF is a multiple of $d \text{ assoc}$. $\square$

The interior- λ hypothesis in (ii) is essential rather than technical. At $\lambda = 0$ or $\lambda = 1$ only one route direction enters the first-order coefficient, and the annihilator of a single direction is three-dimensional rather than two, so vanishing on an open set of priors pins ∇F only to that larger space, not to the span of the constants and ∇assoc .

A.5. Proof of Proposition ADJ

With Q the belief after the first step and $t = (t_0, t_1)$, $t_1 = (1 - \omega)Q(B=1) + \omega r_1$, the target of the second, column j is multiplied by $t_j/Q(B=j)$, so that

$$P(B=1) = t_1,$$

$$P(A=1) = \sum_j Q(A=1, B=j) \frac{t_j}{Q(B=j)}$$

$$= (1 - \omega)Q(A=1) + \omega \sum_j Q(A=1, B=j) \frac{r_j}{Q(B=j)}.$$

In sequence AB , $Q = P^A$, $Q(A=1) = q_1$, and the last sum is $P_{AB}^J(A=1)$, giving

$$P_{AB}^\omega(B=1) - r_1 = (1 - \omega)[P^A(B=1) - r_1], \quad P_{AB}^\omega(A=1) - q_1 = \omega[P_{AB}^J(A=1) - q_1].$$

In sequence BA , by symmetry, $P_{BA}^\omega(A=1) = (1 - \omega) P^B(A=1) + \omega q_1$; subtracting gives the third line. Neither bracket vanishes identically, since

$$P^A(B=1) - r_1 \Big|_{c=0} = r_0 - \beta, \quad P_{AB}^J(A=1) - q_1 = -c \frac{q_0(1 - q_0)(r_0 - \beta)}{\alpha\beta(1 - \alpha)(1 - \beta)} + \mathcal{O}(c^2),$$

the second by Proposition DRF, whose proof also gives $P_{AB}^J(A=1) - P^B(A=1) = \mathcal{O}(c^2)$, hence the coefficient claim. At $c = 0$ a step leaves the other marginal unchanged, so

$$P_{AB}^J(A=1) = q_1, \quad P^B(A=1) = 1 - \alpha, \quad P_{AB}^\omega(A=1) - P_{BA}^\omega(A=1) = (1 - \omega)(\alpha - q_0).$$

For $P^A(B=1) \neq r_1$ the first line gives

$$1 - \omega = \frac{P_{AB}^\omega(B=1) - r_1}{P^A(B=1) - r_1}, \quad \omega = \frac{P^A(B=1) - P_{AB}^\omega(B=1)}{P^A(B=1) - r_1}.$$

□

A.6. Proof of Proposition LAD

Every Jeffrey step, damped or not, multiplies the rows or the columns of the table by constants, so after both steps $P_\sigma^\omega(i, j) = a_i b_j P(i, j)$ for some factors a_0, a_1, b_0, b_1 . Such a rescaling multiplies the association by $a_0 a_1 b_0 b_1$ and leaves the odds ratio unchanged, and $\text{assoc}(P) = c$, so $\text{assoc}(P_\sigma^\omega) = c k_\sigma$ with $k_\sigma = a_0 a_1 b_0 b_1$, and the odds ratio claim of (iii) follows. (i) At $c = 0$ a step on one attribute leaves the other marginal unchanged (Proposition IMM), so sequence AB reaches $q \otimes \beta$ and then $q \otimes t$, and sequence BA reaches $s \otimes r$; subtracting gives the display. A statistic protected for every pair of cues has a differential that annihilates both tangent directions of the independence surface at each of its points (Lemma ASC), so it is constant on the surface and takes the same value at $q \otimes t$ and $s \otimes r$. (ii) At $c = 0$ the factors give $k_{AB} = q_0 q_1 t_0 t_1 / Z$ and $k_{BA} = s_0 s_1 r_0 r_1 / Z$, and $k_{AB} - k_{BA} = (1 - \omega)H / Z$. (iii) Near the independence surface a statistic protected for every pair of cues is $F = F_0 + \text{assoc} \cdot h$ with h smooth, and writing $h = g/m$ with $m = P(A=0)P(A=1)P(B=0)P(B=1)$, the first-order coefficient of $F(P_{AB}^\omega) - F(P_{BA}^\omega)$ is $k_{AB} h(q \otimes t) - k_{BA} h(s \otimes r) = [g(q \otimes t) - g(s \otimes r)] / Z$, since $m(q \otimes t) = q_0 q_1 t_0 t_1$ and $m(s \otimes r) = s_0 s_1 r_0 r_1$. It vanishes for every pair of cues and every ω exactly when g is constant on the independence surface, that is when the differential of F there is a multiple of the differential of assoc / m , which at an independent belief equals the differential of the log odds ratio. □

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used Anthropic's Claude (large language model) in order to draft and refine manuscript prose; propose, prove, and cross-verify Proposition PRO (uniqueness of the protected statistic), which was then independently re-derived and machine-verified by the author; propose bibliographic sources for positioning against neighbouring literatures; and structure the manuscript to conform to the target journal's house style. After using this tool, the author reviewed and edited the content as needed, independently verified the mathematical statements, and takes full responsibility for the content of the published article.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Arrow, K.J., 1973. The theory of discrimination, in: Ashenfelter, O., Rees, A. (Eds.), *Discrimination in Labor Markets*. Princeton University Press, Princeton, NJ, pp. 3–33.

- Asch, S.E., 1946. Forming impressions of personality. *Journal of Abnormal and Social Psychology* 41, 258–290.
- Banerjee, A.V., 1992. A simple model of herd behavior. *The Quarterly Journal of Economics* 107, 797–817. doi:10.2307/2118364.
- Bikhchandani, S., Hirshleifer, D., Welch, I., 1992. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy* 100, 992–1026. doi:10.1086/261849.
- Bohren, J.A., Imas, A., Rosenberg, M., 2019. The dynamics of discrimination: Theory and evidence. *American Economic Review* 109, 3395–3436.
- Bordalo, P., Coffman, K., Gennaioli, N., Shleifer, A., 2016. Stereotypes. *The Quarterly Journal of Economics* 131, 1753–1794. doi:10.1093/qje/qjw029.
- Coate, S., Loury, G.C., 1993. Will affirmative-action policies eliminate negative stereotypes? *American Economic Review* 83, 1220–1240.
- Coffman, K.B., Exley, C.L., Niederle, M., 2021. The role of beliefs in driving gender discrimination. *Management Science* 67, 3551–3569. doi:10.1287/mnsc.2020.3660.
- Cripps, M.W., 2021. Divisible updating. Manuscript, University College London.
- Diaconis, P., Zabell, S.L., 1982. Updating subjective probability. *Journal of the American Statistical Association* 77, 822–830. doi:10.1080/01621459.1982.10477893.
- Domotor, Z., 1980. Probability kinematics and representation of belief change. *Philosophy of Science* 47, 384–403.
- Döring, F., 1999. Why Bayesian psychology is incomplete. *Philosophy of Science* 66, S379–S389.
- Epstein, L.G., 2006. An axiomatic model of non-Bayesian updating. *Review of Economic Studies* 73, 413–436. doi:10.1111/j.1467-937X.2006.00381.x.
- Field, H., 1978. A note on Jeffrey conditionalization. *Philosophy of Science* 45, 361–367.
- Foster, J., Greer, J., Thorbecke, E., 1984. A class of decomposable poverty measures. *Econometrica* 52, 761–766.
- Hawthorne, J., 2004. Three models of sequential belief updating on uncertain evidence. *Journal of Philosophical Logic* 33, 89–123.
- Heckman, J.J., 1998. Detecting discrimination. *Journal of Economic Perspectives* 12, 101–116.
- Hogarth, R.M., Einhorn, H.J., 1992. Order effects in belief updating: The belief-adjustment model. *Cognitive Psychology* 24, 1–55. doi:10.1016/0010-0285(92)90002-J.
- Jeffrey, R.C., 1983. *The Logic of Decision*. 2nd ed., University of Chicago Press, Chicago. First edition 1965.
- Jeffrey, R.C., 2004. *Subjective Probability: The Real Thing*. Cambridge University Press, Cambridge.
- Ortoleva, P., 2012. Modeling the change of paradigm: Non-Bayesian reactions to unexpected news. *American Economic Review* 102, 2410–2436. doi:10.1257/aer.102.6.2410.
- Pettigrew, R., Weisberg, J., 2025. Jeffrey pooling. *Philosophers' Imprint* 25, 1–16. doi:10.3998/phimp.3806.
- Tao, T., 2011. *An Introduction to Measure Theory*. volume 126 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI. Author's version freely available at <https://terrytao.wordpress.com/books/an-introduction-to-measure-theory/>.
- Tversky, A., Kahneman, D., 1992. Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty* 5, 297–323. doi:10.1007/BF00122574.
- Wagner, C.G., 2002. Probability kinematics and commutativity. *Philosophy of Science* 69, 266–278. doi:10.1086/341051.